

Trijet Production - Simplified

(massive quarks)

What changed with respect to the massless note. The quark and the antiquark of flavour f carry the mass m_f (the gluon stays massless). All matrix elements, energy denominators, Fourier transforms, wave functions and cross sections of the massless note are rewritten below with the mass kept everywhere; nothing is expanded in m_f . Setting $m_f = 0$ in every formula gives back the massless note (after the corrections listed in Appendix B). Because the new radial functions depend on m_f , the flavour sum $\sum_f e_f^2$ now sits inside the coordinate integrals, and the factor Q^2 that multiplied the massless cross sections is absorbed into the radial functions. The Wilson-line structures $\mathcal{W}$ do not depend on the quark mass and are unchanged.

1 Setup

The cross section for trijet production is given by

$$\frac{d^3 N}{d^3 k_1 d^3 k_2 d^3 k_3} (2\pi) \delta(q^+ - k_1^+ - k_2^+ - k_3^+) = \frac{1}{2} \int_{w, w'} e^{-iq \cdot (w - w')} \langle \gamma_T^i(Q, q^+, w') | \mathcal{N}_q(k_1) \mathcal{N}_{\bar{q}}(k_2) \mathcal{N}_g(k_3) | \gamma_T^i(Q, q^+, w) \rangle_{q\bar{q}g} \quad (1.1)$$

1.1 Kinematics

The virtual photon with the four momentum as $q^\mu = (q^+, -\frac{Q^2}{2q^+}, q)$. Let us define the kinematics in the following:

Component	Longitudinal momentum fraction	Transverse momentum	Mass
Quark	$\zeta = \frac{k_1^+}{q^+}$	k_1	m_f
Anti Quark	$\eta = 1 - \zeta - \xi$	k_2	m_f
Gluon	$\xi = \frac{k_3^+}{q^+}$	k_3	0

The on-shell light-cone energies are $E_q(k) = \frac{k^2 + m_f^2}{2k^+}$, $E_{\bar{q}}(p) = \frac{p^2 + m_f^2}{2p^+}$ and $E_g(p) = \frac{p^2}{2p^+}$; this is the only place where the mass enters the energy denominators.

1.2 Matrix Elements

The matrix elements that are used in this process are as follows. The mass enters the two-component vertices through $\sigma \cdot k \rightarrow \sigma \cdot k - m_f \sigma^3$ for a quark line and $\sigma \cdot p \rightarrow \sigma \cdot p + m_f \sigma^3$ for an antiquark line (the opposite sign of the mass term of the antiquark spinor); the instantaneous vertex does not change.

1. Virtual photon splitting into a quark and anti quark pair

$$\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) | H_{\gamma \rightarrow q\bar{q}} | \gamma_\lambda(q) \rangle = (2\pi)^3 \delta^{(3)}(q - k - p) \frac{e e_f \delta_{\alpha\beta}}{2\sqrt{2}q^+} \chi_{\lambda_1}^\dagger \left[\frac{2q^\lambda}{q^+} - \sigma^\lambda \frac{\sigma \cdot p + m_f \sigma^3}{p^+} - \frac{\sigma \cdot k - m_f \sigma^3}{k^+} \sigma^\lambda \right] \chi_{\lambda_2} \quad (1.2)$$

rewriting it using the delta function ($k^+ = \zeta q^+$, $p^+ = (1 - \zeta)q^+$, $\tilde{k} = k - \zeta q$), we get

$$\left[\frac{2q^\lambda}{q^+} - \sigma^\lambda \frac{\sigma \cdot (q - k) + m_f \sigma^3}{q^+ - k^+} - \frac{\sigma \cdot k - m_f \sigma^3}{k^+} \sigma^\lambda \right] = \frac{[(2\zeta - 1) \delta^{\lambda j} + i\epsilon^{\lambda j} \sigma^3] \tilde{k}^j + m_f \sigma^3 \sigma^\lambda}{\zeta(1 - \zeta) q^+}$$

where we used $\sigma^i \sigma^j = \delta^{ij} + i\epsilon^{ij} \sigma^3$ and $\sigma^\lambda \sigma^3 = -\sigma^3 \sigma^\lambda$, and

$$\chi_{\lambda_1}^\dagger [(2\zeta - 1) \delta^{\lambda j} + i\epsilon^{\lambda j} \sigma^3] \chi_{\lambda_2} = \delta_{\lambda_1 \lambda_2} [(2\zeta - 1) \delta^{\lambda j} + 2i\epsilon^{\lambda j} \lambda_1], \quad \chi_{\lambda_1}^\dagger \sigma^3 \sigma^\lambda \chi_{\lambda_2} = \delta_{\lambda_2, -\lambda_1} [2\lambda_1 \delta^{\lambda 1} - i\delta^{\lambda 2}] \equiv \mu_{\lambda_1 \lambda_2}^\lambda.$$

The mass term flips the helicity ($\lambda_2 = -\lambda_1$). The matrix element defined above can be rewritten as follows:

$$\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) | H_{\gamma \rightarrow q\bar{q}} | \gamma_\lambda(q) \rangle = (2\pi)^3 \delta^{(3)}(q - k - p) \frac{e e_f \delta_{\alpha\beta}}{2\sqrt{2}q^{+3} \zeta(1 - \zeta)} \left\{ [(2\zeta - 1) \delta^{\lambda j} + 2i\epsilon^{\lambda j} \lambda_1] \delta_{\lambda_1 \lambda_2} \tilde{k}^j + m_f \mu_{\lambda_1 \lambda_2}^\lambda \right\} \quad (1.3)$$

2. Gluon emission from anti - quark

$$\langle q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p') g_{i'}^{\alpha'}(q') | H_{\bar{q}g} | q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) \rangle = -(2\pi)^6 \delta^{(3)}(k - k') \delta^{(3)}(p - p' - q') \delta_{\alpha' \alpha} \delta_{\lambda_1 \lambda_1'} \frac{g t_{\beta \beta'}^{\alpha'}}{2\sqrt{2}q'^+} \chi_{\lambda_2}^\dagger \left[\frac{2q'^{i'}}{q'^+} - \sigma^{i'} \frac{\sigma \cdot p' + m_f \sigma^3}{p'^+} - \frac{\sigma \cdot p + m_f \sigma^3}{p^+} \sigma^{i'} \right] \chi_{\lambda_2'} \quad (1.4)$$

With $p^+ = (1 - \zeta)q^+$, $p'^+ = \eta q^+$, $q'^+ = \xi q^+$ and $\tilde{p} = q' - \frac{\xi}{1-\zeta} p$ the terms proportional to p cancel and

$$\chi_\lambda^\dagger [\dots] \chi_{\lambda_2} = \frac{\tau_{\lambda\lambda_2}^{i'n}(\xi, \eta) \tilde{p}^n + \xi \omega_f \mu_{\lambda\lambda_2}^{i'}}{\xi \eta q^+}, \quad \omega_f = \frac{\xi m_f}{1 - \zeta},$$

because the mass terms combine into $m_f \sigma^3 \sigma^{i'} \left(\frac{1}{p'^+} - \frac{1}{p^+} \right) = \frac{\xi^2 m_f}{(1-\zeta)} \frac{\sigma^3 \sigma^{i'}}{\xi \eta q^+}$. Again the mass term flips the antiquark helicity.

3. Instantaneous contribution (unchanged: the γ^+ in the instantaneous propagator removes the mass)

$$\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(s) g_j^b(p) | H_{\gamma \rightarrow q\bar{q}g} | \gamma \rangle = -(2\pi)^3 \delta^{(3)}(q - k - p - s) \frac{g e e_f \delta_{\lambda_1 \lambda_2} t_{\alpha\beta}^b}{4 \sqrt{q^+ p^+}} \frac{\delta^{ij} + 2i \epsilon^{ij} \lambda_1}{p^+ + s^+} \quad (1.5)$$

1.3 Notations

$$\tilde{k} = k - \zeta q, \quad \tilde{p} = p - \frac{\xi}{1-\zeta} (q - k), \quad \tilde{Q}^2 = \zeta(1 - \zeta) Q^2$$

$$\varepsilon_f^2 = \tilde{Q}^2 + m_f^2 = \zeta(1 - \zeta) Q^2 + m_f^2, \quad \omega_f = \frac{\xi m_f}{1 - \zeta}, \quad E_f^2 = Q^2 + m_f^2 \left(\frac{1}{\zeta} + \frac{1}{\eta} \right), \quad a = \frac{\xi \eta}{\zeta(1 - \zeta)^2}$$

$$\varphi_{\lambda_1 \lambda}^{ij}(\zeta) = \delta_{\lambda_1 \lambda} [(2\zeta - 1) \delta^{ij} + 2i \epsilon^{ij} \lambda_1], \quad \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) = \delta_{\lambda \lambda_2} [(2(1 - \zeta) - \xi) \delta^{mn} + 2i \xi \epsilon^{mn} \lambda]$$

$$\mu_{\lambda \lambda'}^i = \delta_{\lambda', -\lambda} [2\lambda \delta^{i1} - i \delta^{i2}] = \chi_\lambda^\dagger \sigma^3 \sigma^i \chi_{\lambda'} \quad (\text{helicity-flip vertex; } \sum_i \mu_{\lambda \lambda'}^i \mu_{\lambda \lambda'}^{i*} = 2 \delta_{\lambda', -\lambda})$$

Here ε_f is the usual mass-shifted photon scale (not the Levi-Civita symbol ϵ^{ij}). The photon vertex and the antiquark-gluon vertex are

$$\text{photon: } \varphi_{\lambda_1 \lambda}^{ij}(\zeta) \tilde{k}^j + m_f \mu_{\lambda_1 \lambda}^i, \quad \text{gluon off the antiquark: } \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) \tilde{p}^n + \xi \omega_f \mu_{\lambda \lambda_2}^m.$$

The energy denominator is as follows:

$$E_q + E_{\bar{q}} - E_\gamma = \frac{\tilde{k}^2 + \tilde{Q}^2 + m_f^2}{2\zeta(1 - \zeta) q^+} = \frac{\tilde{k}^2 + \varepsilon_f^2}{2\zeta(1 - \zeta) q^+}$$

$$E_{q_1} + E_{\bar{q}_1} + E_{g_1} - E_\gamma = \frac{(1 - \zeta)(\tilde{p}^2 + \omega_f^2)}{2\xi \eta q^+} + \frac{(\tilde{k}^2 + \varepsilon_f^2)}{2\zeta(1 - \zeta) q^+} = \frac{\zeta(1 - \zeta)^2(\tilde{p}^2 + \omega_f^2) + \xi \eta(\tilde{k}^2 + \varepsilon_f^2)}{2\zeta(1 - \zeta) \xi \eta q^+}$$

$$\begin{aligned} E_{q_1} + E_{\bar{q}_1} + E_{g_1} - E_q - E_{\bar{q}} &= \frac{p^2}{2q^+ \xi} + \frac{(q - k - p)^2 + m_f^2}{2q^+ \eta} - \frac{(q - k)^2 + m_f^2}{2q^+(1 - \zeta)} \\ &= \frac{[(1 - \zeta)p - \xi(q - k)]^2}{2\xi(1 - \zeta) \eta q^+} + \frac{m_f^2}{2q^+} \left(\frac{1}{\eta} - \frac{1}{1 - \zeta} \right) = \frac{(1 - \zeta)^2 \tilde{p}^2 + \xi^2 m_f^2}{2\xi(1 - \zeta) \eta q^+} \end{aligned}$$

Simplifying it we get

$$E_{q_1} + E_{\bar{q}_1} + E_{g_1} - E_q - E_{\bar{q}} = \frac{(1 - \zeta)(\tilde{p}^2 + \omega_f^2)}{2\xi \eta q^+}$$

The gluon emission therefore sees the “dead-cone” scale $\omega_f = \xi m_f / (1 - \zeta)$, and when only the three-parton denominator is present (instantaneous term) the total scale is

$$\zeta(1 - \zeta)^2 \omega_f^2 + \xi \eta \varepsilon_f^2 = \zeta(1 - \zeta) \xi \eta E_f^2, \quad E_f^2 = Q^2 + \frac{m_f^2}{\zeta} + \frac{m_f^2}{\eta}.$$

2 Outgoing component of wave function through LCPT

1. Anti quark emits a gluon before the shock wave

Fig. 1: $\bar{q} \rightarrow \bar{q}g$ before the shock wave
Wave functions/1.jpeg

$$|\gamma\rangle_{q\bar{q}g}^{(1)} = |q_2 \bar{q}_2 g_2\rangle \langle q_2 \bar{q}_2 g_2| S |q_1 \bar{q}_1 g_1\rangle \times \frac{\langle q_1 \bar{q}_1 g_1 | H_{\bar{q}g} | q\bar{q} \rangle}{E_{\bar{q}_1} + E_{q_1} + E_{g_1} - E_\gamma} \times \frac{\langle q\bar{q} | H_{\gamma \rightarrow q\bar{q}} | \gamma \rangle}{E_q + E_{\bar{q}} - E_\gamma} \quad (2.1)$$

$$\begin{aligned} |\gamma\rangle_{q\bar{q}g}^{(1)} &= \int d^3 k'' d^3 p'' d^3 r d^3 k' d^3 p' d^3 q' d^3 k d^3 p |q_{\lambda_1}^{\alpha''}(k'') \bar{q}_{\lambda_2}^{\beta''}(p'') g_j^b(r)\rangle \langle q_{\lambda_1}^{\alpha''}(k'') \bar{q}_{\lambda_2}^{\beta''}(p'') g_j^b(r)| S |q_{\lambda_1}^{\alpha'}(k') \bar{q}_{\lambda_2}^{\beta'}(p') g_i^{a'}(q')\rangle \times \\ &\frac{\langle q_{\lambda_1}^{\alpha'}(k') \bar{q}_{\lambda_2}^{\beta'}(p') g_i^{a'}(q') | H_{\bar{q}g} | q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) \rangle}{E_{\bar{q}}(p) + E_q(k') + E_g(q') - E_\gamma} \times \frac{\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) | H_{\gamma \rightarrow q\bar{q}} | \gamma_\lambda(q) \rangle}{E_{\bar{q}}(p) + E_q(k) - E_\gamma(q, Q)} \end{aligned} \quad (2.2)$$

We will first evaluate the matrix elements without scattering (i.e., $S = 1$) [Eq. (4.21) in 2211.04837]. With the massive vertices and energy denominators of Sec. 1,

$$|\gamma_i(Q, q^+, q)\rangle_{q\bar{q}g}^{(1a)} = -\frac{1}{(2\pi)^6} \int d^2\tilde{k} d^2\tilde{p} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, k) \bar{q}_{\lambda_2}^\beta(\eta q^+, q - k - p) g_m^a(\xi q^+, p)\rangle \frac{e e_f g q^+ \zeta(1-\zeta)}{2\sqrt{\xi}(\tilde{k}^2 + \varepsilon_f^2)} \\ \frac{\left[\varphi_{\lambda_1\lambda}^{ij}(\zeta) \tilde{k}^j + m_f \mu_{\lambda_1\lambda}^i \right] \left[\tau_{\lambda\lambda_2}^{mn}(\xi, \eta) \tilde{p}^n + \xi \omega_f \mu_{\lambda\lambda_2}^m \right] t_{\alpha\beta}^a}{\left[\zeta(1-\zeta)^2 (\tilde{p}^2 + \omega_f^2) + \xi \eta (\tilde{k}^2 + \varepsilon_f^2) \right]} \quad (2.3)$$

(a sum over the intermediate antiquark helicity λ is understood). Compared to the massless case: $\tilde{Q}^2 \rightarrow \varepsilon_f^2$, $\tilde{p}^2 \rightarrow \tilde{p}^2 + \omega_f^2$, and each vertex acquires its helicity-flip mass term. The product of the two brackets therefore contains four helicity paths:

$$\underbrace{\varphi \tau \tilde{k}^j \tilde{p}^n}_{\text{no flip}}, \quad \underbrace{\xi \omega_f \varphi \mu \tilde{k}^j}_{\text{flip at the gluon vertex}}, \quad \underbrace{m_f \mu \tau \tilde{p}^n}_{\text{flip at the photon vertex}}, \quad \underbrace{m_f \xi \omega_f \mu \mu}_{\text{two flips}}.$$

The first and the last have $\lambda_2 = \lambda_1$; the two middle ones have $\lambda_2 = -\lambda_1$.

However, Wilson lines can be obtained only if the wave function is expressed in transverse coordinates. Therefore, we perform a Fourier transform to rewrite the wave function in transverse coordinate space. Performing Fourier transform for transverse co-ordinates,

$$|q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(q - k - p) g_m^a(p)\rangle = \int_{x,y,z} e^{ik \cdot x} e^{ip \cdot z} e^{i(q-k-p) \cdot y} |q_{\lambda_1}^\alpha(x) \bar{q}_{\lambda_2}^\beta(y) g_m^a(z)\rangle \quad (2.4)$$

Rewriting k, p in terms of $\tilde{k}$ and $\tilde{p}$ respectively, we get

$$e^{i[\zeta x + \eta y + \xi z] \cdot q} e^{-i(y-z) \cdot \tilde{p}} e^{i[(1-\zeta)x - \eta y - \xi z] \cdot \frac{\tilde{k}}{(1-\zeta)}} = e^{i(\zeta x + \eta y + \xi z) \cdot q} e^{-i(y-z) \cdot \tilde{p}} e^{i(x - \frac{\eta y + \xi z}{(1-\zeta)}) \cdot \tilde{k}} \\ = e^{i w \cdot q} e^{-i Y \cdot \tilde{p}} e^{i R \cdot \tilde{k}}. \quad (2.5)$$

The transverse coordinates of the photon (w) and the intermediate antiquark (y' , before emitting the gluon) are fixed by centre of energy conservation at the two vertices (this is kinematics and does not depend on the mass):

$$w = \zeta x + (1-\zeta) y' \\ = \zeta x + \eta y + \xi z, \quad (2.6)$$

and

$$y' = \frac{\eta y + \xi z}{(1-\zeta)}. \quad (2.7)$$

The relative distances are defined as

$$R = x - y'; \quad Y = y - z$$

where we used

$$(1-\zeta)x - \eta y - \xi z = (1-\zeta)(x - y') = (1-\zeta)R$$

In the massless case we used the following Fourier transform (note the sign of the exponent: with $e^{+ip \cdot Z}$ the result is as written; our phase above is $e^{-i\tilde{p} \cdot Y}$, i.e. $Z = -Y$):

$$\int \frac{d^2k}{2\pi} \frac{d^2p}{2\pi} \frac{k^i p^j}{k^2 [ak^2 + bp^2]} e^{ik \cdot X + ip \cdot Z} = -\frac{X^i Z^j}{Z^2 [bX^2 + aZ^2]} \quad (2.8)$$

With masses the two denominators contain two different scales, ε_f and ω_f , and the transform is no longer a single Bessel function. We need four transforms, one for each helicity path. Define

$$(1-\zeta) \int \frac{d^2\tilde{k}}{2\pi} \frac{d^2\tilde{p}}{2\pi} \frac{e^{i\tilde{k} \cdot R} e^{-i\tilde{p} \cdot Y} \left\{ \tilde{k}^j \tilde{p}^n, \tilde{k}^j, \tilde{p}^n, 1 \right\}}{(\tilde{k}^2 + \varepsilon_f^2) \left[\zeta(1-\zeta)^2 (\tilde{p}^2 + \omega_f^2) + \xi \eta (\tilde{k}^2 + \varepsilon_f^2) \right]} = \left\{ R^j Y^n \mathcal{G}^{(11)}, i R^j \mathcal{G}^{(10)}, -i Y^n \mathcal{G}^{(01)}, \mathcal{G}^{(00)} \right\} \quad (2.9)$$

To evaluate them we use Schwinger parameters,

$$\frac{1}{(\tilde{k}^2 + \varepsilon_f^2) \left[\zeta(1-\zeta)^2 (\tilde{p}^2 + \omega_f^2) + \xi \eta (\tilde{k}^2 + \varepsilon_f^2) \right]} = \frac{1}{\xi \eta} \int_0^\infty du \int_0^\infty dv e^{-(u+v)(\tilde{k}^2 + \varepsilon_f^2) - \frac{v}{\alpha} (\tilde{p}^2 + \omega_f^2)},$$

and the Gaussian integrals

$$\int \frac{d^2\tilde{k}}{2\pi} e^{i\tilde{k} \cdot R - U \tilde{k}^2} \left\{ 1, \tilde{k}^j \right\} = \frac{e^{-R^2/4U}}{2U} \left\{ 1, \frac{iR^j}{2U} \right\}, \quad \int \frac{d^2\tilde{p}}{2\pi} e^{-i\tilde{p} \cdot Y - S \tilde{p}^2} \left\{ 1, \tilde{p}^n \right\} = \frac{e^{-Y^2/4S}}{2S} \left\{ 1, -\frac{iY^n}{2S} \right\},$$

with $U = u+v$ and $S = v/a$. Writing $S = xU/a$ with $x \in (0, 1)$ (x is a Feynman parameter) and using $\int_0^\infty dU U^{\nu-1} e^{-AU-B/U} = 2(B/A)^{\nu/2} K_\nu(2\sqrt{AB})$, every transform becomes a one-dimensional integral:

$$\mathcal{G}^{(\alpha\beta)}(R, Y) = \frac{a^\beta}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x^{1+\beta}} \left(\frac{A_x}{\Delta_x} \right)^{\frac{\alpha+\beta}{2}} K_{\alpha+\beta}(\sqrt{A_x \Delta_x}), \quad \alpha, \beta \in \{0, 1\},$$

$$A_x = \varepsilon_f^2 + \frac{x \omega_f^2}{a} = \zeta(1-\zeta) Q^2 + m_f^2 \left(1 + \frac{x \zeta \xi}{\eta} \right), \quad \Delta_x = R^2 + \frac{a Y^2}{x}. \quad (2.10)$$

Explicitly,

$$\mathcal{G}^{(11)} = \frac{a}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x^2} \frac{A_x}{\Delta_x} K_2(\sqrt{A_x \Delta_x}), \quad \mathcal{G}^{(10)} = \frac{1}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x} \sqrt{\frac{A_x}{\Delta_x}} K_1(\sqrt{A_x \Delta_x}),$$

$$\mathcal{G}^{(01)} = \frac{a}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x^2} \sqrt{\frac{A_x}{\Delta_x}} K_1(\sqrt{A_x \Delta_x}), \quad \mathcal{G}^{(00)} = \frac{1}{2\zeta(1-\zeta)} \int_0^1 \frac{dx}{x} K_0(\sqrt{A_x \Delta_x}). \quad (2.11)$$

The end point $x = 1$ is the three-parton configuration: $A_1 \Delta_1 = E_f^2 \mathcal{D}^2$, with $\mathcal{D}$ defined below. For $m_f = 0$, $A_x = \tilde{Q}^2$ does not depend on x ; substituting $s^2 = R^2 + aY^2/x$ and using $\frac{d}{ds} \left[\frac{K_1(\tilde{Q}s)}{s} \right] = -\frac{\tilde{Q} K_2(\tilde{Q}s)}{s}$ gives back the massless closed form,

$$\mathcal{G}^{(11)} \Big|_{m_f=0} = \frac{\tilde{Q} K_1(\tilde{Q}\sqrt{R^2 + aY^2})}{\zeta(1-\zeta) Y^2 \sqrt{R^2 + aY^2}} = \frac{1}{Y^2} \frac{Q K_1(Q \mathcal{D})}{\mathcal{D}}, \quad \mathcal{G}^{(10)}, \mathcal{G}^{(01)}, \mathcal{G}^{(00)} \text{ appear multiplied by } m_f.$$

where $a = \frac{\xi \eta}{[\zeta(1-\zeta)^2]}$ and K_ν is the modified Bessel function of the second kind. Now, implementing all these steps in the outgoing component of wave function, we get

$$|\gamma_i(Q, q^+, q)\rangle_{q\bar{q}g}^{(1a)} = -\frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\zeta}{\sqrt{\xi}} \delta^{(2)}(w-c) \Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{G}] t_{\alpha\beta}^a \quad (2.12)$$

with the massive regular vertex function (the radial functions are evaluated at (R, Y))

$$\Psi_{\lambda_1 \lambda_2}^{im}[F] = \sum_\lambda \left[\varphi_{\lambda_1 \lambda}^{ij}(\zeta) \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) R^j Y^n F^{(11)} + i \xi \omega_f \varphi_{\lambda_1 \lambda}^{ij}(\zeta) \mu_{\lambda \lambda_2}^m R^j F^{(10)} \right. \\ \left. - i m_f \mu_{\lambda_1 \lambda}^i \tau_{\lambda \lambda_2}^{mn}(\xi, \eta) Y^n F^{(01)} + m_f \xi \omega_f \mu_{\lambda_1 \lambda}^i \mu_{\lambda \lambda_2}^m F^{(00)} \right], \quad (2.13)$$

where

$$\mathcal{D}^2(x, y, z, \zeta, \xi) = \zeta(1-\zeta) R^2 + \frac{\xi \eta Y^2}{(1-\zeta)} = \zeta \eta (x-y)^2 + \zeta \xi (x-z)^2 + \eta \xi (y-z)^2$$

and $c = \zeta x + \eta y + \xi z$ is the centre of energy of trijet system. For $m_f = 0$ only the first term survives and $\Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{G}] \rightarrow \frac{R^j Y^n}{Y^2} \varphi_{\lambda_1 \lambda}^{ij} \tau_{\lambda \lambda_2}^{mn} \frac{Q K_1(Q \mathcal{D})}{\mathcal{D}}$, which is the massless result. Attaching Wilson lines, we get [(4.23)?]

$$|\gamma_i(Q, q^+, q)\rangle_{q\bar{q}g}^{(1a)} = -\frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\zeta}{\sqrt{\xi}} \delta^{(2)}(w-c) \times \\ \Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{G}](R, Y) \times [U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} \quad (2.14)$$

2. Anti quark emits a gluon after the shock wave

Fig. 2: $\bar{q} \rightarrow \bar{q}g$ after the shock wave
Wave functions/2.jpeg

$$|\gamma\rangle_{q\bar{q}g}^{(2)} = |q_2 \bar{q}_2 g_2\rangle \langle q_2 \bar{q}_2 g_2| S |q_1 \bar{q}_1 g_1\rangle \times \frac{\langle q_1 \bar{q}_1 g_1 | H_{\bar{q}g} | q \bar{q} \rangle}{E_{\bar{q}_1} + E_{q_1} - E_{q_2} - E_{\bar{q}_2} - E_{g_2}} \times \frac{\langle q \bar{q} | H_{\gamma \rightarrow q \bar{q}} | \gamma \rangle}{E_q + E_{\bar{q}} - E_\gamma} \quad (2.15)$$

$$|\gamma\rangle_{q\bar{q}g}^{(2)} = \int d^3 k'' d^3 p'' d^3 r d^3 k' d^3 p' d^3 k d^3 p |q_{\lambda_1'}^{\alpha''}(k'') \bar{q}_{\lambda_2'}^{\beta''}(p'') g_j^b(r)\rangle \frac{\langle q_{\lambda_1'}^{\alpha''}(k'') \bar{q}_{\lambda_2'}^{\beta''}(p'') g_j^b(r) | H_{\bar{q}g} | q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p') \rangle}{E_q(k'') + E_{\bar{q}}(p'') - E_q(k') - E_{\bar{q}}(p') - E_g(r)} \times \\ \langle q_{\lambda_1'}^{\alpha'}(k') \bar{q}_{\lambda_2'}^{\beta'}(p') | S | q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) \rangle \times \frac{\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p) | H_{\gamma \rightarrow q \bar{q}} | \gamma_\lambda(q) \rangle}{E_{\bar{q}}(p) + E_q(k) - E_\gamma(q, Q)} \quad (2.16)$$

The only difference between the first and the second wave function component is the energy denominator and the Wilson line structure. The changes are as follows:

$$\frac{1}{E_{q\bar{q}g} - E_\gamma} \rightarrow \frac{1}{E_{q\bar{q}} - E_{q\bar{q}g}}$$

Equivalently,

$$\frac{1}{\zeta(1-\zeta)^2(\tilde{p}^2 + \omega_f^2) + \xi\eta(\tilde{k}^2 + \varepsilon_f^2)} \rightarrow -\frac{1}{\zeta(1-\zeta)^2(\tilde{p}^2 + \omega_f^2)}$$

The $\tilde{k}$ and $\tilde{p}$ integrals now factorise. The Fourier transforms which we will be using here are the following:

$$\int \frac{d^2\tilde{k}}{2\pi} \frac{\{\tilde{k}^j, 1\}}{\tilde{k}^2 + \varepsilon_f^2} e^{i\tilde{k}\cdot R} = \left\{ \frac{iR^j}{R} \varepsilon_f K_1(\varepsilon_f R), K_0(\varepsilon_f R) \right\}, \quad \int \frac{d^2\tilde{p}}{2\pi} \frac{\{\tilde{p}^n, 1\}}{\tilde{p}^2 + \omega_f^2} e^{-i\tilde{p}\cdot Y} = \left\{ -\frac{iY^n}{Y} \omega_f K_1(\omega_f Y), K_0(\omega_f Y) \right\}$$

(for $m_f = 0$: $\omega_f K_1(\omega_f Y)/Y \rightarrow 1/Y^2$, which is the massless $-iY^n/Y^2$). With the same normalisation as Eq. (2.9),

$$(1-\zeta) \int \frac{d^2\tilde{k}}{2\pi} \frac{d^2\tilde{p}}{2\pi} \frac{e^{i\tilde{k}\cdot R} e^{-i\tilde{p}\cdot Y} \{\tilde{k}^j \tilde{p}^n, \tilde{k}^j, \tilde{p}^n, 1\}}{(\tilde{k}^2 + \varepsilon_f^2) \zeta(1-\zeta)^2(\tilde{p}^2 + \omega_f^2)} = \left\{ R^j Y^n \mathcal{A}^{(11)}, iR^j \mathcal{A}^{(10)}, -iY^n \mathcal{A}^{(01)}, \mathcal{A}^{(00)} \right\}$$

with

$$\begin{aligned} \mathcal{A}^{(11)} &= \frac{\varepsilon_f K_1(\varepsilon_f R)}{\zeta(1-\zeta)R} \frac{\omega_f K_1(\omega_f Y)}{Y}, & \mathcal{A}^{(10)} &= \frac{\varepsilon_f K_1(\varepsilon_f R)}{\zeta(1-\zeta)R} K_0(\omega_f Y), \\ \mathcal{A}^{(01)} &= \frac{K_0(\varepsilon_f R)}{\zeta(1-\zeta)} \frac{\omega_f K_1(\omega_f Y)}{Y}, & \mathcal{A}^{(00)} &= \frac{K_0(\varepsilon_f R)}{\zeta(1-\zeta)} K_0(\omega_f Y). \end{aligned} \quad (2.17)$$

The Wilson line will not have any component of gluon because the gluons are not getting scattered as the gluon is emitted after the scattering.

$$[U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} \rightarrow [V(x) V^\dagger(y') t^a - t^a]_{\alpha\beta}$$

Substituting all these changes in eq. (2.14), we get

$$\begin{aligned} |\gamma(w)\rangle_{q\bar{q}g}^{(2)} &= + \frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\zeta}{\sqrt{\xi}} \delta^{(2)}(w-c) \times \\ &\Psi_{\lambda_1 \lambda_2}^{im}[\mathcal{A}](R, Y) \times [V(x) V^\dagger(y') t^a - t^a]_{\alpha\beta} \end{aligned} \quad (2.18)$$

In the massless case $\mathcal{A}^{(11)} = \frac{1}{Y^2} \frac{Q^2 K_1(\bar{Q}R)}{\bar{Q}R}$ is exactly $\mathcal{G}^{(11)}$ with $\mathcal{D} \rightarrow \mathcal{D}(x, y', y', \zeta, \xi) = \sqrt{\zeta(1-\zeta)} |x-y'| = \frac{\bar{Q}R}{Q}$ (keeping the prefactor $1/Y^2$), which is the formal substitution $(b) = (a)|_{y,z \rightarrow y'}$ of the paper. With masses the after-shock-wave functions are the genuinely different functions (2.17): the gluon is emitted from an on-shell antiquark and sees only the dead-cone scale ω_f . We keep the notation “ $[y, z \rightarrow y']$ ” for this term, meaning: Wilson lines at y' and $\mathcal{G} \rightarrow \mathcal{A}$. The important property survives: at short gluon-antiquark distance both functions have the same singular behaviour,

$$Y^2 \mathcal{G}^{(11)} \xrightarrow{Y \rightarrow 0} \frac{\varepsilon_f K_1(\varepsilon_f R)}{\zeta(1-\zeta)R} \xleftarrow{Y \rightarrow 0} Y^2 \mathcal{A}^{(11)}, \quad Y^2 \mathcal{G}^{(01)} \xrightarrow{Y \rightarrow 0} \frac{K_0(\varepsilon_f R)}{\zeta(1-\zeta)} \xleftarrow{Y \rightarrow 0} Y^2 \mathcal{A}^{(01)},$$

and $\mathcal{G}^{(10)}, \mathcal{G}^{(00)}$ and $\mathcal{A}^{(10)}, \mathcal{A}^{(00)}$ both grow only like $\ln(1/Y)$ with the same coefficient, so the ultraviolet ($Y \rightarrow 0$) cancellation between Eq. (2.14) and Eq. (2.18) works as in the massless case.

3. Instantaneous contribution (Anti quark emission of gluon when an instantaneous vertex acts before the shockwave)

Fig. 5: instantaneous,
antiquark side
Wave functions/5.png

$$|\gamma\rangle_{q\bar{q}g}^{(1 \text{ inst.})} = |q\bar{q}g\rangle \langle q\bar{q}g| S |q\bar{q}g\rangle \frac{\langle q\bar{q}g| H_{\gamma \rightarrow q\bar{q}g} |\gamma\rangle}{E_{q\bar{q}g} - E_\gamma} \quad (2.19)$$

Using the relevant matrix we get (In momentum space - without Wilson lines). The vertex is mass independent; the mass enters only through the three-parton energy denominator:

$$\begin{aligned} |\gamma\rangle_{q\bar{q}g}^{(1 \text{ inst.})} &= + \frac{1}{(2\pi)^6} \int d^2\tilde{k} d^2\tilde{p} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, k) \bar{q}_{\lambda_2}^\beta(\eta q^+, q-k-p) g_m^a(\xi q^+, p)\rangle \frac{e e_f g q^+ \zeta}{2\sqrt{\xi}} \\ &\frac{(\delta^{im} + 2i\epsilon^{im}\lambda_1) \delta_{\lambda_1 \lambda_2} t_{\alpha\beta}^a}{\tilde{k}^2 + \varepsilon_f^2 + \frac{\zeta(1-\zeta)^2}{\xi\eta}(\tilde{p}^2 + \omega_f^2)} \end{aligned} \quad (2.20)$$

The Fourier transform is a four-dimensional one (rescale $\tilde{p} = \sqrt{a} p'$, so that $\tilde{k}^2 + \tilde{p}^2/a$ becomes a 4D square and $R^2 + aY^2 = \mathcal{D}^2/[\zeta(1-\zeta)]$), and the constant term is exactly $\varepsilon_f^2 + \omega_f^2/a = \zeta(1-\zeta)E_f^2$:

$$(1-\zeta) \int \frac{d^2\tilde{k}}{2\pi} \frac{d^2\tilde{p}}{2\pi} \frac{e^{i\tilde{k}\cdot R} e^{-i\tilde{p}\cdot Y}}{\zeta(1-\zeta)^2(\tilde{p}^2 + \omega_f^2) + \xi\eta(\tilde{k}^2 + \varepsilon_f^2)} = \frac{E_f K_1(E_f \mathcal{D})}{\mathcal{D}} \equiv \mathcal{G}_E. \quad (2.21)$$

So for this single-denominator term the mass effect is exactly $Q \rightarrow E_f$. Performing Fourier transform and inserting $S-1$ component, we get the following [(4.28)]

$$\begin{aligned} |\gamma\rangle_{q\bar{q}g}^{(1\text{ inst.})} &= + \frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\zeta\eta\sqrt{\xi}}{(1-\zeta)} \\ &\quad \delta^{(2)}(w-c) (\delta^{im} + 2i\epsilon^{im}\lambda_1) \delta_{\lambda_1\lambda_2} \\ &\quad \frac{E_f K_1(E_f \mathcal{D})}{\mathcal{D}} [U^{ab}(z)V(x)t^b V^\dagger(y) - t^a]_{\alpha\beta} \end{aligned} \quad (2.22)$$

This has the same Wilson line structure as the regular i.e., Eq. (2.14). There is no instantaneous term after the shock wave (the gluon emission after the shock wave is a single vertex), so the instantaneous term has no $[y, z \rightarrow y']$ counterpart; in the massless note this was expressed by the factor $\frac{Y^2}{R\cdot Y}$ of the effective vertex, which vanishes for $y, z \rightarrow y'$. Now, we have all the wave functions corresponding to anti quark emission. With masses the regular and instantaneous pieces have different radial functions, so instead of the massless effective vertex Φ we collect them into the massive effective vertex function

$$\mathbb{M}_{\lambda_1\lambda_2}^{im}(x, y, z, \zeta, \xi) = \Psi_{\lambda_1\lambda_2}^{im}[\mathcal{G}](R, Y) - \delta_{\lambda_1\lambda_2} \frac{\xi\eta}{(1-\zeta)} (\delta^{im} + 2i\epsilon^{im}\lambda_1) \mathcal{G}_E, \quad \mathbb{M}_{\lambda_1\lambda_2}^{im}\Big|_{\text{after}} = \Psi_{\lambda_1\lambda_2}^{im}[\mathcal{A}](R, Y). \quad (2.23)$$

For $m_f = 0$: $\mathbb{M}_{\lambda_1\lambda_2}^{im} = \frac{R^j Y^n}{Y^2} \Phi_{\lambda_1\lambda_2}^{ijmn} \frac{QK_1(Q\mathcal{D})}{\mathcal{D}}$ with the massless effective vertex

$$\Phi_{\lambda_1\lambda_2}^{ijmn}(x, y, z, \zeta, \xi) = \varphi_{\lambda_1\lambda}^{ij}(\zeta) \tau_{\lambda\lambda_2}^{mn}(\xi, \eta) - \delta_{\lambda_1\lambda_2} \delta^{nj} \frac{\xi\eta}{(1-\zeta)} (\delta^{im} + 2i\epsilon^{im}\lambda_1) \frac{Y^2}{R\cdot Y}.$$

All four anti quark emissions are then written as a single expression

$$\begin{aligned} |\gamma^i(w)\rangle_{q\bar{q}g} &= - \frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\zeta}{\sqrt{\xi}} \delta^{(2)}(w-c) \\ &\quad \left[\mathbb{M}_{\lambda_1\lambda_2}^{im}(x, y, z, \zeta, \xi) [U^{ab}(z)V(x)t^b V^\dagger(y) - t^a]_{\alpha\beta} - \mathbb{M}_{\lambda_1\lambda_2}^{im}\Big|_{\text{after}} [V(x)V^\dagger(y')t^a - t^a]_{\alpha\beta} \right] \end{aligned} \quad (2.24)$$

4. Quark emits a gluon before the shock wave

Fig. 3: $q \rightarrow qg$ before the shock wave
Wave functions/3.jpeg

When we move from anti quark emission to quark emission, we will have to perform the following changes from anti quark wave functions:

- $\zeta \leftrightarrow \eta$
- $x \leftrightarrow y$
- $\alpha \leftrightarrow \beta$
- $\lambda_1 \leftrightarrow \lambda_2$
- Change overall sign
- $y' \rightarrow x' = \frac{\zeta x + \xi z}{\zeta + \xi}$
- $R \rightarrow y - x' = -\tilde{R}$
- $Y \rightarrow x - z = X$
- $\varphi_{\lambda_1\lambda}^{ij}(\zeta) \rightarrow \varphi_{\lambda_2\lambda}^{ij*}(\eta) = -\varphi_{\lambda\lambda_2}^{ij}(\zeta + \xi)$
- $\tau_{\lambda\lambda_2}^{mn}(\xi, \eta) \rightarrow \tau_{\lambda\lambda_1}^{mn*}(\xi, \zeta) = \tau_{\lambda_1\lambda}^{nm}(\xi, \zeta)$
- mass vertices: $m_f \mu_{\lambda_1\lambda}^i \rightarrow m_f \mu_{\lambda\lambda_2}^i$ (photon) and $\xi\omega_f \mu_{\lambda\lambda_2}^m \rightarrow \xi\tilde{\omega}_f \mu_{\lambda_1\lambda}^m$ (gluon), with $\tilde{\omega}_f = \frac{\xi m_f}{1-\eta}$
- $\varepsilon_f^2 \rightarrow \tilde{\varepsilon}_f^2 = \eta(1-\eta)Q^2 + m_f^2$, $a \rightarrow \tilde{a} = \frac{\xi\zeta}{\eta(1-\eta)^2}$, $A_x \rightarrow \tilde{A}_x = \eta(1-\eta)Q^2 + m_f^2 \left(1 + \frac{x\eta\xi}{\zeta}\right)$, $\Delta_x \rightarrow \tilde{\Delta}_x = \tilde{R}^2 + \frac{\tilde{a}X^2}{x}$
- $\mathcal{G}^{(\alpha\beta)}(R, Y) \rightarrow \tilde{\mathcal{G}}^{(\alpha\beta)}(\tilde{R}, X)$ and $\mathcal{A}^{(\alpha\beta)}(R, Y) \rightarrow \tilde{\mathcal{A}}^{(\alpha\beta)}(\tilde{R}, X)$ (same formulas with the replacements above and $\zeta(1-\zeta) \rightarrow \eta(1-\eta)$)

- $\mathcal{D}(x, y, z, \zeta, \xi) \rightarrow \mathcal{D}(y, x, z, \eta, \xi) = \mathcal{D}(x, y, z, \zeta, \xi)$ and E_f are symmetric, so $\mathcal{G}_E$ is the same function in both channels
- after the shock wave (massless) $\mathcal{D} \rightarrow \sqrt{\eta(1-\eta)} \tilde{R}$, i.e. $\tilde{Q} \rightarrow \sqrt{\eta(1-\eta)} Q$

The vertex-level check of these rules (they are not only a relabelling: the mass terms were obtained independently from the Dirac spinors of the quark line) is described in Appendix C.

$$|\gamma\rangle_{q\bar{q}g}^{(3)} = |q_2\bar{q}_2g_2\rangle \langle q_2\bar{q}_2g_2| S |q_1\bar{q}_1g_1\rangle \times \frac{\langle q_1\bar{q}_1g_1| H_{qg} |q\bar{q}\rangle}{E_{q_1} + E_{q_1} + E_{g_1} - E_\gamma} \times \frac{\langle q\bar{q}| H_{\gamma \rightarrow q\bar{q}} |\gamma\rangle}{E_q + E_{\bar{q}} - E_\gamma} \quad (2.25)$$

$$|\gamma\rangle_{q\bar{q}g}^{(3)} = \int d^3k'' d^3p'' d^3r d^3k' d^3p' d^3q' d^3k d^3p |q_{\lambda_1}^{\alpha''}(k'') \bar{q}_{\lambda_2}^{\beta''}(p'') g_j^b(r)\rangle \langle q_{\lambda_1}^{\alpha''}(k'') \bar{q}_{\lambda_2}^{\beta''}(p'') g_j^b(r)| S |q_{\lambda_1}^{\alpha'}(k') \bar{q}_{\lambda_2}^{\beta'}(p') g_{i'}^a(q')\rangle \times \frac{\langle q_{\lambda_1}^{\alpha'}(k') \bar{q}_{\lambda_2}^{\beta'}(p') g_{i'}^a(q')| H_{qg} |q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p)\rangle}{E_{\bar{q}}(p') + E_q(k') + E_g(q') - E_\gamma} \times \frac{\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p)| H_{\gamma \rightarrow q\bar{q}} |\gamma_\lambda(q)\rangle}{E_{\bar{q}}(p) + E_q(k) - E_\gamma(q, Q)} \quad (2.26)$$

In momentum space the relative momenta are $K = (q - k_2) - (1 - \eta) q$ (parent quark relative to the photon) and $\tilde{p}_q = k_3 - \frac{\xi}{1-\eta} (q - k_2)$ (gluon relative to the parent quark), conjugate to $\tilde{R}$ and $-X$; the energy denominators are those of Sec. 1 with $\zeta \leftrightarrow \eta$, i.e. $\frac{K^2 + \tilde{\varepsilon}_f^2}{2\eta(1-\eta)q^+}$ and $\frac{\eta(1-\eta)^2(\tilde{p}_q^2 + \tilde{\omega}_f^2) + \xi\zeta(K^2 + \tilde{\varepsilon}_f^2)}{2\eta(1-\eta)\xi\zeta q^+}$. The result is

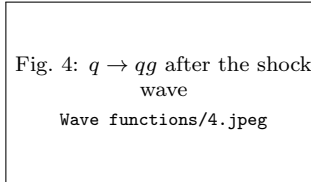
$$|\gamma^i(w)\rangle_{q\bar{q}g}^{(3)} = \frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\eta}{\sqrt{\xi}} \delta^{(2)}(w - c) \times \left[\tilde{\Psi}_{\lambda_1\lambda_2}^{im}[\tilde{\mathcal{G}}](\tilde{R}, X) \times [U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} \right] \quad (2.27)$$

with

$$\tilde{\Psi}_{\lambda_1\lambda_2}^{im}[F] = \sum_\lambda \left[\tau_{\lambda_1\lambda}^{nm}(\xi, \zeta) \varphi_{\lambda\lambda_2}^{ij}(\zeta + \xi) \tilde{R}^j X^n F^{(11)} + i \xi \tilde{\omega}_f \mu_{\lambda_1\lambda}^m \varphi_{\lambda\lambda_2}^{ij}(\zeta + \xi) \tilde{R}^j F^{(10)} - i m_f \tau_{\lambda_1\lambda}^{nm}(\xi, \zeta) \mu_{\lambda\lambda_2}^i X^n F^{(01)} + m_f \xi \tilde{\omega}_f \mu_{\lambda_1\lambda}^m \mu_{\lambda\lambda_2}^i F^{(00)} \right]. \quad (2.28)$$

For $m_f = 0$, $\tilde{\Psi}[\tilde{\mathcal{G}}] \rightarrow \frac{\tilde{R}^j X^n}{X^2} \tau_{\lambda_1\lambda}^{nm}(\xi, \zeta) \varphi_{\lambda\lambda_2}^{ij}(\xi + \zeta) \frac{QK_1(QD)}{D}$, the massless Eq. (eq:5) of the original note (L230).

5. Quark emits a gluon after the shock wave



$$|\gamma\rangle_{q\bar{q}g}^{(4)} = |q_2\bar{q}_2g_2\rangle \langle q_2\bar{q}_2g_2| S |q_1\bar{q}_1g_1\rangle \times \frac{\langle q_1\bar{q}_1g_1| H_{qg} |q\bar{q}\rangle}{E_{q_1} + E_{q_1} - E_{q_2} - E_{\bar{q}_2} - E_{g_2}} \times \frac{\langle q\bar{q}| H_{\gamma \rightarrow q\bar{q}} |\gamma\rangle}{E_q + E_{\bar{q}} - E_\gamma} \quad (2.29)$$

$$|\gamma\rangle_{q\bar{q}g}^{(4)} = \int d^3k'' d^3p'' d^3r d^3k' d^3p' d^3k d^3p |q_{\lambda_1}^{\alpha''}(k'') \bar{q}_{\lambda_2}^{\beta''}(p'') g_j^b(r)\rangle \frac{\langle q_{\lambda_1}^{\alpha''}(k'') \bar{q}_{\lambda_2}^{\beta''}(p'') g_j^b(r)| H_{qg} |q_{\lambda_1}^{\alpha'}(k') \bar{q}_{\lambda_2}^{\beta'}(p')\rangle}{E_q(k') + E_{\bar{q}}(p') - E_q(k'') - E_{\bar{q}}(p'') - E_g(r)} \times \frac{\langle q_{\lambda_1}^{\alpha'}(k') \bar{q}_{\lambda_2}^{\beta'}(p')| S |q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p)\rangle}{E_{\bar{q}}(p) + E_q(k) - E_\gamma(q, Q)} \times \frac{\langle q_{\lambda_1}^\alpha(k) \bar{q}_{\lambda_2}^\beta(p)| H_{\gamma \rightarrow q\bar{q}} |\gamma_\lambda(q)\rangle}{E_{\bar{q}}(p) + E_q(k) - E_\gamma(q, Q)} \quad (2.30)$$

Substituting the relations we get

$$|\gamma^i(w)\rangle_{q\bar{q}g}^{(4)} = -\frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\eta}{\sqrt{\xi}} \delta^{(2)}(w - c) \times \tilde{\Psi}_{\lambda_1\lambda_2}^{im}[\tilde{\mathcal{A}}](\tilde{R}, X) \times [t^a V(x') V^\dagger(y) - t^a]_{\alpha\beta} \quad (2.31)$$

with

$$\begin{aligned} \tilde{\mathcal{A}}^{(11)} &= \frac{\tilde{\varepsilon}_f K_1(\tilde{\varepsilon}_f \tilde{R})}{\eta(1-\eta) \tilde{R}} \frac{\tilde{\omega}_f K_1(\tilde{\omega}_f X)}{X}, & \tilde{\mathcal{A}}^{(10)} &= \frac{\tilde{\varepsilon}_f K_1(\tilde{\varepsilon}_f \tilde{R})}{\eta(1-\eta) \tilde{R}} K_0(\tilde{\omega}_f X), \\ \tilde{\mathcal{A}}^{(01)} &= \frac{K_0(\tilde{\varepsilon}_f \tilde{R})}{\eta(1-\eta)} \frac{\tilde{\omega}_f K_1(\tilde{\omega}_f X)}{X}, & \tilde{\mathcal{A}}^{(00)} &= \frac{K_0(\tilde{\varepsilon}_f \tilde{R}) K_0(\tilde{\omega}_f X)}{\eta(1-\eta)}. \end{aligned} \quad (2.32)$$

For $m_f = 0$: $\tilde{\mathcal{A}}^{(11)} = \frac{1}{X^2} \frac{Q^2 K_1(\tilde{Q}_q \tilde{R})}{\tilde{Q}_q \tilde{R}}$ with $\tilde{Q}_q^2 = \eta(1-\eta) Q^2$.

Fig. 6: instantaneous, quark side
Wave functions/6.png

6. Instantaneous contribution (Quark emission of gluon when an instantaneous vertex acts before the shockwave)

$$|\gamma\rangle_{q\bar{q}g}^{(3\text{ inst.})} = |q\bar{q}g\rangle \langle q\bar{q}g| S |q\bar{q}g\rangle \frac{\langle q\bar{q}g| H_{\gamma \rightarrow q\bar{q}g} |\gamma\rangle}{E_{q\bar{q}g} - E_\gamma} \quad (2.33)$$

Using the transformations from anti quark to quark with the insertions of Wilson lines (the Fourier transform is Eq. (2.21) with $1 - \zeta \rightarrow 1 - \eta$), we get

$$\begin{aligned} |\gamma\rangle_{q\bar{q}g}^{(3\text{ inst.})} &= -\frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\zeta \eta \sqrt{\xi}}{(1-\eta)} \\ &\quad \delta^{(2)}(w-c) (\delta^{im} - 2i \epsilon^{im} \lambda_2) \delta_{\lambda_1 \lambda_2} \\ &\quad \frac{E_f K_1(E_f \mathcal{D})}{\mathcal{D}} [U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} \end{aligned} \quad (2.34)$$

Now, we will write the quark emissions as how we wrote the anti quark section. We will use the massive effective vertex function

$$\tilde{\mathbb{M}}_{\lambda_1 \lambda_2}^{im}(x, y, z, \eta, \xi) = \tilde{\Psi}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{G}}](\tilde{R}, X) - \delta_{\lambda_1 \lambda_2} \frac{\zeta \xi}{(\xi + \zeta)} (\delta^{im} - 2i \epsilon^{im} \lambda_1) \mathcal{G}_E, \quad \tilde{\mathbb{M}}_{\lambda_1 \lambda_2}^{im}|_{\text{after}} = \tilde{\Psi}_{\lambda_1 \lambda_2}^{im}[\tilde{\mathcal{A}}](\tilde{R}, X), \quad (2.35)$$

$$\begin{aligned} |\gamma^i(w)\rangle_{q\bar{q}g} &= \frac{e e_f g q^+}{2(2\pi)^4} \int_{x,y,z} \int_0^1 d\zeta \int_0^{1-\zeta} d\xi |q_{\lambda_1}^\alpha(\zeta q^+, x) \bar{q}_{\lambda_2}^\beta(\eta q^+, y) g_m^a(\xi q^+, z)\rangle \frac{\eta}{\sqrt{\xi}} \delta^{(2)}(w-c) \times \\ &\quad \left[\tilde{\mathbb{M}}_{\lambda_1 \lambda_2}^{im}(x, y, z, \eta, \xi) \times [U^{ab}(z) V(x) t^b V^\dagger(y) - t^a]_{\alpha\beta} - \tilde{\mathbb{M}}_{\lambda_1 \lambda_2}^{im}|_{\text{after}} [t^a V(x') V^\dagger(y) - t^a]_{\alpha\beta} \right] \end{aligned} \quad (2.36)$$

For $m_f = 0$, $\tilde{\mathbb{M}}_{\lambda_1 \lambda_2}^{im} = \frac{\tilde{R}^j X^n}{X^2} \tilde{\Phi}_{\lambda_1 \lambda_2}^{ijmn} \frac{Q K_1(Q \mathcal{D})}{\mathcal{D}}$ with the massless effective vertex

$$\tilde{\Phi}_{\lambda_1 \lambda_2}^{ijmn}(x, y, z, \eta, \xi) = \tau_{\lambda_1 \lambda}^{nm}(\xi, \zeta) \varphi_{\lambda \lambda_2}^{ij}(\xi + \zeta) - \delta_{\lambda_1 \lambda_2} \delta^{nj} \frac{\zeta \xi}{(\xi + \zeta)} (\delta^{im} - 2i \epsilon^{im} \lambda_1) \frac{X^2}{\tilde{R} \cdot X}.$$

3 Cross section

3.1 Direct contributions - Anti quark contribution

The cross section for trijet production is given by [(5.2)]

$$\frac{d^3 N}{d^3 k_1 d^3 k_2 d^3 k_3} (2\pi) \delta(q^+ - k_1^+ - k_2^+ - k_3^+) = \frac{1}{2} \int_{w,w'} e^{-iq \cdot (w-w')} \langle \gamma_T^i(Q, q^+, w') | \mathcal{N}_q(k_1) \mathcal{N}_{\bar{q}}(k_2) \mathcal{N}_g(k_3) | \gamma_T^i(Q, q^+, w) \rangle_{q\bar{q}g} \quad (3.1)$$

Inserting Eq. (2.24) in the amplitude and in the complex conjugate amplitude (barred coordinates) gives

$$\begin{aligned} \frac{d^3 N^{\text{direct}}}{d^3 k_1 d^3 k_2 d^3 k_3} &= \frac{e^2 g^2 \zeta^2 C_F N_c}{8(2\pi)^{12} \xi q^{+2}} \delta(q^+ - k_1^+ - k_2^+ - k_3^+) \int_{x,y,z,\bar{x},\bar{y},\bar{z}} e^{-i k_1 \cdot (x-\bar{x})} e^{-i k_2 \cdot (y-\bar{y})} e^{-i k_3 \cdot (z-\bar{z})} \sum_f e_f^2 \\ &\quad \left[\mathcal{H}[\mathcal{G}; \bar{\mathcal{G}}] \mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) - \mathcal{H}[\mathcal{A}; \bar{\mathcal{G}}] \mathcal{W}(x, y', y', \bar{x}, \bar{y}, \bar{z}) \right. \\ &\quad \left. - \mathcal{H}[\mathcal{G}; \bar{\mathcal{A}}] \mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') + \mathcal{H}[\mathcal{A}; \bar{\mathcal{A}}] \mathcal{W}(x, y', y', \bar{x}, \bar{y}', \bar{y}') \right] \end{aligned} \quad (3.2)$$

where

$$\mathcal{H}[F; \bar{F}] = \sum_{\lambda_1, \lambda_2} \sum_{l,r} \mathbb{M}_{\lambda_1 \lambda_2}^{lr}[F](x, y, z) \mathbb{M}_{\lambda_1 \lambda_2}^{lr*}[\bar{F}](\bar{x}, \bar{y}, \bar{z}),$$

$\mathbb{M}[\mathcal{G}]$ is the full vertex function (2.23) (regular + instantaneous) and $\mathbb{M}[\mathcal{A}] = \mathbb{M}|_{\text{after}}$ (regular only, i.e. $\mathcal{G}_E \rightarrow 0$). Barred functions are evaluated at $(\bar{R}, \bar{Y})$, $\bar{F}^{(\alpha\beta)} = F^{(\alpha\beta)}(\bar{R}, \bar{Y})$, $\bar{\mathcal{G}}_E = \mathcal{G}_E(\bar{x}, \bar{y}, \bar{z})$. Compared to the massless formula the factor Q^2 in front is now inside the radial functions ($\frac{Q^2 K_1 K_1}{\mathcal{D}^2} = \frac{Q K_1}{\mathcal{D}} \frac{Q K_1}{\mathcal{D}}$) and the flavour sum is inside the integral. We write $\mathcal{H}$ in parts, exactly as in the massless case, plus the new helicity-flip pieces (index convention: $i \leftrightarrow R$, $m \leftrightarrow Y$, $j \leftrightarrow \bar{R}$, $n \leftrightarrow \bar{Y}$; $A \cdot B$ is the transverse scalar product).

- Regular $\times$ Regular

$$\begin{aligned}\mathcal{H}_{\text{reg} \times \text{reg}} = & R^i Y^m \bar{R}^j \bar{Y}^n \left\{ 8 [\eta^2 + (1 - \zeta)^2] [\zeta^2 + (1 - \zeta)^2] \delta^{ij} \delta^{mn} - 8 \xi (2\zeta - 1) (2(1 - \zeta) - \xi) \epsilon^{ij} \epsilon^{mn} \right\} F^{(11)} \bar{F}^{(11)} \\ & + 8 m_f^2 [\eta^2 + (1 - \zeta)^2] (Y \cdot \bar{Y}) F^{(01)} \bar{F}^{(01)} + 8 \xi^2 \omega_f^2 [\zeta^2 + (1 - \zeta)^2] (R \cdot \bar{R}) F^{(10)} \bar{F}^{(10)} + 8 m_f^2 \xi^2 \omega_f^2 F^{(00)} \bar{F}^{(00)} \\ & - 8 \zeta \eta m_f \xi \omega_f \left[(R \cdot Y) F^{(11)} \bar{F}^{(00)} + (\bar{R} \cdot \bar{Y}) F^{(00)} \bar{F}^{(11)} + (R \cdot \bar{Y}) F^{(10)} \bar{F}^{(01)} + (Y \cdot \bar{R}) F^{(01)} \bar{F}^{(10)} \right]\end{aligned}\quad (3.3)$$

The first line is the massless result (helicity conserving at both vertices); the second line is the square of the paths with one helicity flip (m_f^2 times the massless gluon-vertex factor $\eta^2 + (1 - \zeta)^2$, and $\xi^2 \omega_f^2$ times the massless photon-vertex factor $\zeta^2 + (1 - \zeta)^2$) and of the double flip; the third line is the interference between the paths with the same final helicities (no-flip with double-flip, and the two single-flip paths).

- Regular $\times$ Instantaneous + h.c.

$$\mathcal{H}_{\text{reg} \times \text{inst}} = - \frac{8 \zeta \xi \eta^2}{(1 - \zeta)} \left[(R \cdot Y) F^{(11)} \bar{\mathcal{G}}_E + (\bar{R} \cdot \bar{Y}) \mathcal{G}_E \bar{F}^{(11)} \right] + \frac{8 \xi^2 \eta m_f \omega_f}{(1 - \zeta)} \left[F^{(00)} \bar{\mathcal{G}}_E + \mathcal{G}_E \bar{F}^{(00)} \right] \quad (3.4)$$

(only the helicity-conserving paths interfere with the instantaneous term, which has $\lambda_2 = \lambda_1$). The first bracket follows from

$$\sum_{l,r} \varphi_{\lambda_1 \lambda_1}^{li}(\zeta) \tau_{\lambda_1 \lambda_1}^{rm}(\xi, \eta) (\delta^{lr} + 2i \epsilon^{lr} \lambda_1)^* = 4 \zeta \eta (\delta^{im} - 2i \epsilon^{im} \lambda_1) \Rightarrow \sum_{\lambda_1} \rightarrow 8 \zeta \eta \delta^{im}.$$

- Instantaneous $\times$ Instantaneous

$$\mathcal{H}_{\text{inst} \times \text{inst}} = 8 \left[\frac{\xi \eta}{(1 - \zeta)} \right]^2 \mathcal{G}_E \bar{\mathcal{G}}_E \quad (3.5)$$

- For the terms where one side is “after the shock wave” use $F = \mathcal{A}$ on that side and drop its $\mathcal{G}_E$ (so $\mathcal{H}[\mathcal{A}; \bar{\mathcal{G}}]$ keeps only the $\bar{\mathcal{G}}_E$ terms of the Regular $\times$ Instantaneous line, and $\mathcal{H}[\mathcal{A}; \bar{\mathcal{A}}]$ is Regular $\times$ Regular only).

Massless limit. For $m_f = 0$: $F^{(11)} = \frac{1}{Y^2} \frac{Q K_1(Q\mathcal{D})}{\mathcal{D}}$, $\mathcal{G}_E = \frac{Q K_1(Q\mathcal{D})}{\mathcal{D}}$ and the mass lines drop. With the common factor $\frac{R^i Y^m \bar{R}^j \bar{Y}^n}{Y^2 \bar{Y}^2} \frac{Q^2 K_1 K_1}{\mathcal{D} \bar{\mathcal{D}}}$ taken out as in the massless note,

$$\Phi_{\lambda_1 \lambda}^{lirm} \Phi_{\lambda_1 \lambda}^{ljrn*} \Big|_{\text{reg} \times \text{inst}} = - \frac{8 \zeta \xi \eta^2}{(1 - \zeta)} \delta^{im} \delta^{jn} \left[\frac{Y^2}{R \cdot Y} + \frac{\bar{Y}^2}{\bar{R} \cdot \bar{Y}} \right], \quad \Phi_{\lambda_1 \lambda}^{lirm} \Phi_{\lambda_1 \lambda}^{ljrn*} \Big|_{\text{inst} \times \text{inst}} = 8 \delta^{im} \delta^{jn} \left[\frac{\xi \eta}{(1 - \zeta)} \right]^2 \frac{Y^2}{R \cdot Y} \frac{\bar{Y}^2}{\bar{R} \cdot \bar{Y}}, \quad (3.6)$$

and the Regular $\times$ Regular part is the massless one unchanged. (These two lines replace the corresponding massless expressions, see Appendix B.)

$\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z})$ is the Wilson line structure (independent of the quark mass). This has 4 terms:

- $\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z})$

$$\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) = S_{q\bar{q}g q\bar{q}g}(x, y, z, \bar{x}, \bar{y}, \bar{z}) - S_{q\bar{q}g}(x, y, z) - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) + 1 \quad (3.7)$$

where the Wilson line averaging is given by

$$S_{q\bar{q}g q\bar{q}g}(x, y, z, \bar{x}, \bar{y}, \bar{z}) = \frac{1}{C_F N_c} \langle [U^\dagger(\bar{z}) U(z)]^{ab} \text{Tr} [V(\bar{y}) t^a V^\dagger(\bar{x}) V(x) t^b V^\dagger(y)] \rangle$$

$$S_{q\bar{q}g}(x, y, z) = \frac{1}{C_F N_c} \langle U^{ab}(z) \text{Tr} [V(x) t^a V^\dagger(y) t^b] \rangle$$

At large N_c limit, we get

$$S_{q\bar{q}g q\bar{q}g}(x, y, z, \bar{x}, \bar{y}, \bar{z}) = Q(x, z, \bar{z}, \bar{x}) Q(z, y, \bar{y}, \bar{z})$$

$$S_{q\bar{q}g}(x, y, z) = S(x, z) S(z, y)$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) = S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z})$$

Using Gaussian approximation to reduce quadrupole in to product of dipoles, we get

$$\begin{aligned}Q(x, z, \bar{z}, \bar{x}) = & \exp \left[-\frac{C_F}{2} (\Gamma(x - z) + \Gamma(\bar{z} - \bar{x})) \right] - \frac{F(x, z; \bar{z}, \bar{x})}{F(x, \bar{z}; z, \bar{x})} \left\{ \exp \left[-\frac{C_F}{2} (\Gamma(x - z) + \Gamma(\bar{z} - \bar{x})) \right] \right. \\ & \left. - \exp \left[-\frac{C_F}{2} (\Gamma(x - \bar{x}) + \Gamma(\bar{z} - z)) \right] \right\}\end{aligned}\quad (3.8)$$

where

$$\exp \left[-\frac{C_F}{2} (\Gamma(x - z) + \Gamma(\bar{z} - \bar{x})) \right] = S(x, z) S(\bar{z}, \bar{x})$$

$$\exp \left[-\frac{C_F}{2} (\Gamma(x - \bar{x}) + \Gamma(\bar{z} - z)) \right] = S(x, \bar{x}) S(\bar{z}, z)$$

So,

$$Q(x, z, \bar{z}, \bar{x}) = S(x, z) S(\bar{z}, \bar{x}) - \frac{F(x, z; \bar{z}, \bar{x})}{F(x, \bar{z}; z, \bar{x})} [S(x, z) S(\bar{z}, \bar{x}) - S(x, \bar{x}) S(\bar{z}, z)] \quad (3.9)$$

Similarly,

$$Q(z, y, \bar{y}, \bar{z}) = S(z, y) S(\bar{y}, \bar{z}) - \frac{F(z, y; \bar{y}, \bar{z})}{F(z, \bar{y}; y, \bar{z})} [S(z, y) S(\bar{y}, \bar{z}) - S(z, \bar{z}) S(\bar{y}, y)] \quad (3.10)$$

$$\bullet \mathcal{W}(x, y', y', \bar{x}, \bar{y}, \bar{z})$$

$$\mathcal{W}(x, y', y', \bar{x}, \bar{y}, \bar{z}) = S_{q\bar{q}g q\bar{q}g}(x, y', y', \bar{x}, \bar{y}, \bar{z}) - S_{q\bar{q}g}(x, y', y') - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) + 1 \quad (3.11)$$

where the Wilson line averaging is given by

$$S_{q\bar{q}g q\bar{q}g}(x, y', y', \bar{x}, \bar{y}, \bar{z}) = \frac{1}{C_F N_c} \langle [U^\dagger(\bar{z}) U(y')]^{ab} \text{Tr} [V(\bar{y}) t^a V^\dagger(\bar{x}) V(x) t^b V^\dagger(y')] \rangle$$

$$S_{q\bar{q}g}(x, y', y') = \frac{1}{C_F N_c} \langle U^{ab}(y') \text{Tr} [V(x) t^a V^\dagger(y') t^b] \rangle$$

At large N_c limit, we get

$$S_{q\bar{q}g q\bar{q}g}(x, y', y', \bar{x}, \bar{y}, \bar{z}) = Q(x, y', \bar{z}, \bar{x}) Q(y', y', \bar{y}, \bar{z}) = Q(x, y', \bar{z}, \bar{x}) S(\bar{y}, \bar{z})$$

$$S_{q\bar{q}g}(x, y', y') = S(x, y') S(y', y') = S(x, y')$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) = S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z})$$

Using Gaussian approximation,

$$Q(x, y', \bar{z}, \bar{x}) = S(x, y') S(\bar{z}, \bar{x}) - \frac{F(x, y'; \bar{z}, \bar{x})}{F(x, \bar{z}; y', \bar{x})} [S(x, y') S(\bar{z}, \bar{x}) - S(x, \bar{x}) S(\bar{z}, y')] \quad (3.12)$$

$$\bullet \mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}')$$

$$\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = S_{q\bar{q}g q\bar{q}g}(x, y, z, \bar{x}, \bar{y}', \bar{y}') - S_{q\bar{q}g}(x, y, z) - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') + 1 \quad (3.13)$$

where the Wilson line averaging is given by

$$S_{q\bar{q}g q\bar{q}g}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = \frac{1}{C_F N_c} \langle [U^\dagger(\bar{y}') U(z)]^{ab} \text{Tr} [V(\bar{y}) t^a V^\dagger(\bar{x}) V(x) t^b V^\dagger(y)] \rangle$$

$$S_{q\bar{q}g}(x, y, z) = \frac{1}{C_F N_c} \langle U^{ab}(z) \text{Tr} [V(x) t^a V^\dagger(y) t^b] \rangle$$

At large N_c limit, we get

$$S_{q\bar{q}g q\bar{q}g}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = Q(x, z, \bar{y}', \bar{x}) Q(z, y, \bar{y}', \bar{y}') = Q(x, z, \bar{y}', \bar{x}) S(z, y)$$

$$S_{q\bar{q}g}(x, y, z) = S(x, z) S(z, y)$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) S(\bar{y}', \bar{y}') = S(\bar{y}', \bar{x})$$

Using Gaussian approximation,

$$Q(x, z, \bar{y}', \bar{x}) = S(x, z) S(\bar{y}', \bar{x}) - \frac{F(x, z; \bar{y}', \bar{x})}{F(x, \bar{y}'; z, \bar{x})} [S(x, z) S(\bar{y}', \bar{x}) - S(x, \bar{x}) S(\bar{y}', z)] \quad (3.14)$$

$$\bullet \mathcal{W}(x, y', y', \bar{x}, \bar{y}', \bar{y}')$$

$$\mathcal{W}(x, y', y', \bar{x}, \bar{y}', \bar{y}') = S_{q\bar{q}g q\bar{q}g}(x, y', y', \bar{x}, \bar{y}', \bar{y}') - S_{q\bar{q}g}(x, y', y') - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') + 1 \quad (3.15)$$

where the Wilson line averaging is given by

$$S_{q\bar{q}g q\bar{q}g}(x, y', y', \bar{x}, \bar{y}', \bar{y}') = \frac{1}{C_F N_c} \langle [U^\dagger(\bar{y}') U(y')]^{ab} \text{Tr} [V(\bar{y}') t^a V^\dagger(\bar{x}) V(x) t^b V^\dagger(y')] \rangle$$

$$S_{q\bar{q}g}(x, y', y') = \frac{1}{C_F N_c} \langle U^{ab}(y') \text{Tr} [V(x) t^a V^\dagger(y') t^b] \rangle$$

At large N_c limit, we get

$$S_{q\bar{q}g q\bar{q}g}(x, y', y', \bar{x}, \bar{y}', \bar{y}') = Q(x, y', \bar{y}', \bar{x}) Q(y', y', \bar{y}', \bar{y}') = Q(x, y', \bar{y}', \bar{x})$$

$$S_{q\bar{q}g}(x, y', y') = S(x, y') S(y', y') = S(x, y')$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) S(\bar{y}', \bar{y}') = S(\bar{y}', \bar{x})$$

Using Gaussian approximation,

$$Q(x, y', \bar{y}', \bar{x}) = S(x, y') S(\bar{y}', \bar{x}) - \frac{F(x, y'; \bar{y}', \bar{x})}{F(x, \bar{y}'; y', \bar{x})} [S(x, y') S(\bar{y}', \bar{x}) - S(x, \bar{x}) S(\bar{y}', y')] \quad (3.16)$$

and

$$\mathcal{D}^2(x, y, z, \zeta, \xi) = \zeta (1 - \zeta) R^2 + \frac{\xi \eta Y^2}{(1 - \zeta)} = \zeta \eta (x - y)^2 + \zeta \xi (x - z)^2 + \eta \xi (y - z)^2$$

3.2 Direct contribution - Quark contributions

We need not explicitly derive the direct contribution for quarks. Instead we do the following changes from the cross section of the anti quark sector

$$\zeta \rightarrow \eta, R \rightarrow \tilde{R}, Y \rightarrow X, y' \rightarrow x', x \leftrightarrow y, \quad \varepsilon_f \rightarrow \tilde{\varepsilon}_f, \omega_f \rightarrow \tilde{\omega}_f, \mathcal{G} \rightarrow \tilde{\mathcal{G}}, \mathcal{A} \rightarrow \tilde{\mathcal{A}}$$

and use the quark-emission vertex function $\tilde{\mathbb{M}}$ of Eq. (2.35):

$$\begin{aligned} \frac{d^3 N^{\text{direct}}}{d^3 k_1 d^3 k_2 d^3 k_3} &= \frac{e^2 g^2 \eta^2 C_F N_c}{8 (2\pi)^{12} \xi q^{+2}} \delta(q^+ - k_1^+ - k_2^+ - k_3^+) \int_{x,y,z,\bar{x},\bar{y},\bar{z}} e^{-i k_1 \cdot (x-\bar{x})} e^{-i k_2 \cdot (y-\bar{y})} e^{-i k_3 \cdot (z-\bar{z})} \sum_f e_f^2 \\ &\quad \left[\tilde{\mathcal{H}}[\tilde{\mathcal{G}}; \tilde{\mathcal{G}}] \mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z}) - \tilde{\mathcal{H}}[\tilde{\mathcal{A}}; \tilde{\mathcal{G}}] \mathcal{W}(y, x', x', \bar{y}, \bar{x}, \bar{z}) \right. \\ &\quad \left. - \tilde{\mathcal{H}}[\tilde{\mathcal{G}}; \tilde{\mathcal{A}}] \mathcal{W}(y, x, z, \bar{y}, \bar{x}', \bar{x}') + \tilde{\mathcal{H}}[\tilde{\mathcal{A}}; \tilde{\mathcal{A}}] \mathcal{W}(y, x', x', \bar{y}, \bar{x}', \bar{x}') \right] \end{aligned} \quad (3.17)$$

where $\tilde{\mathcal{H}}[F; \bar{F}] = \sum_{\lambda_1 \lambda_2} \sum_{l,r} \tilde{\mathbb{M}}_{\lambda_1 \lambda_2}^{lr}[F](x, y, z) \tilde{\mathbb{M}}_{\lambda_1 \lambda_2}^{lr*}[\bar{F}](\bar{x}, \bar{y}, \bar{z})$ (after the shock wave: $\mathcal{G}_E \rightarrow 0$). We write it in parts (index convention $i \leftrightarrow \tilde{R}$, $m \leftrightarrow X$, $j \leftrightarrow \tilde{R}$, $n \leftrightarrow \bar{X}$):

- Regular $\times$ Regular

$$\begin{aligned} \tilde{\mathcal{H}}_{\text{reg} \times \text{reg}} &= \tilde{R}^i X^m \tilde{R}^j \bar{X}^n \left\{ 8 [\zeta^2 + (1-\eta)^2] [\eta^2 + (1-\eta)^2] \delta^{ij} \delta^{mn} - 8 \xi (2\eta - 1) (2(1-\eta) - \xi) \epsilon^{ij} \epsilon^{mn} \right\} F^{(11)} \bar{F}^{(11)} \\ &\quad + 8 m_f^2 [\zeta^2 + (1-\eta)^2] (X \cdot \bar{X}) F^{(01)} \bar{F}^{(01)} + 8 \xi^2 \tilde{\omega}_f^2 [\eta^2 + (1-\eta)^2] (\tilde{R} \cdot \bar{R}) F^{(10)} \bar{F}^{(10)} + 8 m_f^2 \xi^2 \tilde{\omega}_f^2 F^{(00)} \bar{F}^{(00)} \\ &\quad + 8 \zeta \eta m_f \xi \tilde{\omega}_f \left[(\tilde{R} \cdot X) F^{(11)} \bar{F}^{(00)} + (\tilde{R} \cdot \bar{X}) F^{(00)} \bar{F}^{(11)} + (\tilde{R} \cdot \bar{X}) F^{(10)} \bar{F}^{(01)} + (X \cdot \tilde{R}) F^{(01)} \bar{F}^{(10)} \right] \end{aligned} \quad (3.18)$$

(the relative sign of the last line with respect to the antiquark case comes from $R \rightarrow -\tilde{R}$.)

- Regular $\times$ Instantaneous + h.c.

$$\tilde{\mathcal{H}}_{\text{reg} \times \text{inst}} = + \frac{8 \zeta^2 \xi \eta}{(1-\eta)} \left[(\tilde{R} \cdot X) F^{(11)} \bar{\mathcal{G}}_E + (\tilde{R} \cdot \bar{X}) \mathcal{G}_E \bar{F}^{(11)} \right] + \frac{8 \xi^2 \zeta m_f \tilde{\omega}_f}{(1-\eta)} \left[F^{(00)} \bar{\mathcal{G}}_E + \mathcal{G}_E \bar{F}^{(00)} \right] \quad (3.19)$$

using $\sum_{l,r} \tau_{\lambda_1 \lambda_1}^{mr}(\xi, \zeta) \varphi_{\lambda_1 \lambda_1}^{li}(\zeta + \xi) (\delta^{lr} - 2i\epsilon^{lr} \lambda_1)^* = -4 \zeta \eta (\delta^{im} + 2i\epsilon^{im} \lambda_1)$.

- Instantaneous $\times$ Instantaneous

$$\tilde{\mathcal{H}}_{\text{inst} \times \text{inst}} = 8 \left[\frac{\xi \zeta}{(1-\eta)} \right]^2 \mathcal{G}_E \bar{\mathcal{G}}_E \quad (3.20)$$

Here $F = \tilde{\mathcal{G}}$ or $\tilde{\mathcal{A}}$ evaluated at $(\tilde{R}, X)$ and $\bar{F}$ at $(\tilde{R}, \bar{X})$. **Massless limit:** with $\frac{\tilde{R}^i X^m \tilde{R}^j \bar{X}^n}{X^2 \bar{X}^2} \frac{Q^2 K_1 K_1}{\mathcal{D} \mathcal{D}}$ taken out,

$$\tilde{\Phi} \tilde{\Phi}^* \Big|_{\text{reg} \times \text{inst}} = + \frac{8 \zeta^2 \xi \eta}{(1-\eta)} \delta^{im} \delta^{jn} \left[\frac{X^2}{\tilde{R} \cdot X} + \frac{\bar{X}^2}{\tilde{R} \cdot \bar{X}} \right], \quad \tilde{\Phi} \tilde{\Phi}^* \Big|_{\text{inst} \times \text{inst}} = 8 \delta^{im} \delta^{jn} \left[\frac{\xi \zeta}{(1-\eta)} \right]^2 \frac{X^2}{\tilde{R} \cdot X} \frac{\bar{X}^2}{\tilde{R} \cdot \bar{X}}.$$

$\mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z})$ is the Wilson line structure (independent of the quark mass). This has 4 terms:

- $\mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z})$

$$\mathcal{W}(y, x, z, \bar{y}, \bar{x}, \bar{z}) = S_{q\bar{q}g q\bar{q}g}(y, x, z, \bar{y}, \bar{x}, \bar{z}) - S_{q\bar{q}g}(y, x, z) - S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}, \bar{z}) + 1 \quad (3.21)$$

where the Wilson line averaging is given by

$$S_{q\bar{q}g q\bar{q}g}(y, x, z, \bar{y}, \bar{x}, \bar{z}) = \frac{1}{C_F N_c} \langle [U^\dagger(\bar{z}) U(z)]^{ab} \text{Tr} [V(\bar{x}) t^a V^\dagger(\bar{y}) V(y) t^b V^\dagger(x)] \rangle$$

$$S_{q\bar{q}g}(y, x, z) = \frac{1}{C_F N_c} \langle U^{ab}(z) \text{Tr} [V(y) t^a V^\dagger(x) t^b] \rangle$$

At large N_c limit, we get

$$S_{q\bar{q}g q\bar{q}g}(y, x, z, \bar{y}, \bar{x}, \bar{z}) = Q(y, z, \bar{z}, \bar{y}) Q(z, x, \bar{x}, \bar{z})$$

$$S_{q\bar{q}g}(y, x, z) = S(y, z) S(z, x)$$

$$S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}, \bar{z}) = S(\bar{z}, \bar{y}) S(\bar{x}, \bar{z})$$

Using Gaussian approximation,

$$Q(y, z, \bar{z}, \bar{y}) = S(y, z) S(\bar{z}, \bar{y}) - \frac{F(y, z; \bar{z}, \bar{y})}{F(y, \bar{z}; z, \bar{y})} [S(y, z) S(\bar{z}, \bar{y}) - S(y, \bar{y}) S(\bar{z}, z)] \quad (3.22)$$

$$Q(z, x, \bar{x}, \bar{z}) = S(z, x) S(\bar{x}, \bar{z}) - \frac{F(z, x; \bar{x}, \bar{z})}{F(z, \bar{x}; x, \bar{z})} [S(z, x) S(\bar{x}, \bar{z}) - S(z, \bar{z}) S(\bar{x}, x)] \quad (3.23)$$

- $\mathcal{W}(y, x', x', \bar{y}, \bar{x}, \bar{z})$

$$\mathcal{W}(y, x', x', \bar{y}, \bar{x}, \bar{z}) = S_{q\bar{q}g\ q\bar{q}g}(y, x', x', \bar{y}, \bar{x}, \bar{z}) - S_{q\bar{q}g}(y, x', x') - S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}, \bar{z}) + 1 \quad (3.24)$$

where the Wilson line averaging is given by

$$S_{q\bar{q}g\ q\bar{q}g}(y, x', x', \bar{y}, \bar{x}, \bar{z}) = \frac{1}{C_F N_c} \langle [U^\dagger(\bar{z})U(x')]^{ab} \text{Tr} [V(\bar{x})t^a V^\dagger(\bar{y})V(y)t^b V^\dagger(x')] \rangle$$

$$S_{q\bar{q}g}(y, x', x') = \frac{1}{C_F N_c} \langle U^{ab}(x') \text{Tr} [V(y)t^a V^\dagger(x')t^b] \rangle$$

At large N_c limit, we get

$$S_{q\bar{q}g\ q\bar{q}g}(y, x', x', \bar{y}, \bar{x}, \bar{z}) = Q(y, x', \bar{z}, \bar{y}) Q(x', x', \bar{x}, \bar{z}) = Q(y, x', \bar{z}, \bar{y}) S(\bar{x}, \bar{z})$$

$$S_{q\bar{q}g}(y, x', x') = S(y, x') S(x', x') = S(y, x')$$

$$S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}, \bar{z}) = S(\bar{z}, \bar{y}) S(\bar{x}, \bar{z})$$

Using Gaussian approximation,

$$Q(y, x', \bar{z}, \bar{y}) = S(y, x') S(\bar{z}, \bar{y}) - \frac{F(y, x'; \bar{z}, \bar{y})}{F(y, \bar{z}; x', \bar{y})} [S(y, x') S(\bar{z}, \bar{y}) - S(y, \bar{y}) S(\bar{z}, x')] \quad (3.25)$$

- $\mathcal{W}(y, x, z, \bar{y}, \bar{x}', \bar{x}')$

$$\mathcal{W}(y, x, z, \bar{y}, \bar{x}', \bar{x}') = S_{q\bar{q}g\ q\bar{q}g}(y, x, z, \bar{y}, \bar{x}', \bar{x}') - S_{q\bar{q}g}(y, x, z) - S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}', \bar{x}') + 1 \quad (3.26)$$

where the Wilson line averaging is given by

$$S_{q\bar{q}g\ q\bar{q}g}(y, x, z, \bar{y}, \bar{x}', \bar{x}') = \frac{1}{C_F N_c} \langle [U^\dagger(\bar{x}')U(z)]^{ab} \text{Tr} [V(\bar{x})t^a V^\dagger(\bar{y})V(y)t^b V^\dagger(x)] \rangle$$

$$S_{q\bar{q}g}(y, x, z) = \frac{1}{C_F N_c} \langle U^{ab}(z) \text{Tr} [V(y)t^a V^\dagger(x)t^b] \rangle$$

At large N_c limit, we get

$$S_{q\bar{q}g\ q\bar{q}g}(y, x, z, \bar{y}, \bar{x}', \bar{x}') = Q(y, z, \bar{x}', \bar{y}) Q(z, x, \bar{x}', \bar{x}') = Q(y, z, \bar{x}', \bar{y}) S(z, x)$$

$$S_{q\bar{q}g}(y, x, z) = S(y, z) S(z, x)$$

$$S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}', \bar{x}') = S(\bar{x}', \bar{y}) S(\bar{x}', \bar{x}') = S(\bar{x}', \bar{y})$$

Using Gaussian approximation,

$$Q(y, z, \bar{x}', \bar{y}) = S(y, z) S(\bar{x}', \bar{y}) - \frac{F(y, z; \bar{x}', \bar{y})}{F(y, \bar{x}'; z, \bar{y})} [S(y, z) S(\bar{x}', \bar{y}) - S(y, \bar{y}) S(\bar{x}', z)] \quad (3.27)$$

- $\mathcal{W}(y, x', x', \bar{y}, \bar{x}', \bar{x}')$

$$\mathcal{W}(y, x', x', \bar{y}, \bar{x}', \bar{x}') = S_{q\bar{q}g\ q\bar{q}g}(y, x', x', \bar{y}, \bar{x}', \bar{x}') - S_{q\bar{q}g}(y, x', x') - S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}', \bar{x}') + 1 \quad (3.28)$$

where the Wilson line averaging is given by

$$S_{q\bar{q}g\ q\bar{q}g}(y, x', x', \bar{y}, \bar{x}', \bar{x}') = \frac{1}{C_F N_c} \langle [U^\dagger(\bar{x}')U(x')]^{ab} \text{Tr} [V(\bar{x}')t^a V^\dagger(\bar{y})V(y)t^b V^\dagger(x')] \rangle$$

$$S_{q\bar{q}g}(y, x', x') = \frac{1}{C_F N_c} \langle U^{ab}(x') \text{Tr} [V(y)t^a V^\dagger(x')t^b] \rangle$$

At large N_c limit, we get

$$S_{q\bar{q}g\ q\bar{q}g}(y, x', x', \bar{y}, \bar{x}', \bar{x}') = Q(y, x', \bar{x}', \bar{y}) Q(x', x', \bar{x}', \bar{x}') = Q(y, x', \bar{x}', \bar{y})$$

$$S_{q\bar{q}g}(y, x', x') = S(y, x') S(x', x') = S(y, x')$$

$$S_{q\bar{q}g}^\dagger(\bar{y}, \bar{x}', \bar{x}') = S(\bar{x}', \bar{y}) S(\bar{x}', \bar{x}') = S(\bar{x}', \bar{y})$$

Using Gaussian approximation,

$$Q(y, x', \bar{x}', \bar{y}) = S(y, x') S(\bar{x}', \bar{y}) - \frac{F(y, x'; \bar{x}', \bar{y})}{F(y, \bar{x}'; x', \bar{y})} [S(y, x') S(\bar{x}', \bar{y}) - S(y, \bar{y}) S(\bar{x}', x')] \quad (3.29)$$

also,

$$\mathcal{D}^2(y, x, z, \eta, \xi) = \eta(1 - \eta) \tilde{R}^2 + \frac{\xi \zeta X^2}{(1 - \eta)} = \eta \zeta (y - x)^2 + \eta \xi (y - z)^2 + \zeta \xi (x - z)^2$$

and

$$R = x - y'; \bar{R} = \bar{x} - \bar{y}'; Y = y - z; \bar{Y} = \bar{y} - \bar{z}; \tilde{R} = x' - y; X = x - z; \bar{X} = \bar{x} - \bar{z}; \bar{\tilde{R}} = \bar{x}' - \bar{y}$$

3.3 Interference contributions

This section contains 8 regular contributions and also their corresponding instantaneous contributions. However, out of 8 regular contributions, only 4 contributions are genuine and the other 4 are its complex conjugate terms. So we compute the genuine contributions (quark emission in the amplitude, Eq. (2.36), times antiquark emission in the complex conjugate amplitude, Eq. (2.24)) and add them with the c.c.

$$\begin{aligned} \frac{d^3 N^{\text{cross}}}{d^3 k_1 d^3 k_2 d^3 k_3} &= -\frac{e^2 g^2 \eta \zeta C_F N_c}{8 (2\pi)^{12} \xi q^{+2}} \delta(q^+ - k_1^+ - k_2^+ - k_3^+) \times 2 \Re \int_{x,y,z,\bar{x},\bar{y},\bar{z}} \\ &\quad e^{-i k_1 \cdot (x-\bar{x})} e^{-i k_2 \cdot (y-\bar{y})} e^{-i k_3 \cdot (z-\bar{z})} \sum_f e_f^2 \\ &\quad \left[\mathcal{H}^\times [\tilde{\mathcal{G}}; \bar{\mathcal{G}}] \mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) - \mathcal{H}^\times [\tilde{\mathcal{A}}; \bar{\mathcal{G}}] \mathcal{W}(x', y, x', \bar{x}, \bar{y}, \bar{z}) \right. \\ &\quad \left. - \mathcal{H}^\times [\tilde{\mathcal{G}}; \bar{\mathcal{A}}] \mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') + \mathcal{H}^\times [\tilde{\mathcal{A}}; \bar{\mathcal{A}}] \mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}') \right] \end{aligned} \quad (3.30)$$

where $\mathcal{H}^\times[\tilde{F}; \bar{F}] = \sum_{\lambda_1 \lambda_2} \sum_{l,r} \tilde{\mathbb{M}}_{\lambda_1 \lambda_2}^{lr}[\tilde{F}](x, y, z) \mathbb{M}_{\lambda_1 \lambda_2}^{lr*}[\bar{F}](\bar{x}, \bar{y}, \bar{z})$, with $\tilde{F}$ at $(\tilde{R}, X)$ and $\bar{F}$ at $(\bar{R}, \bar{Y})$ (after the shock wave: $\mathcal{G}_E \rightarrow 0$ on that side). We write it in parts (index convention $i \leftrightarrow \tilde{R}$, $m \leftrightarrow X$, $j \leftrightarrow \bar{R}$, $n \leftrightarrow \bar{Y}$):

- Regular $\times$ Regular

$$\begin{aligned} \mathcal{H}_{\text{reg} \times \text{reg}}^\times &= \tilde{R}^i X^m \bar{R}^j \bar{Y}^n \left\{ 8 [1 - \xi - 2\zeta\eta] [\xi(1 - \xi) + 2\zeta\eta] \delta^{ij} \delta^{mn} - 8\xi [1 - 2\zeta - \xi]^2 \epsilon^{ij} \epsilon^{mn} \right\} \tilde{F}^{(11)} \bar{F}^{(11)} \\ &\quad + 8m_f^2 (\xi + 2\zeta\eta) (X \cdot \bar{Y}) \tilde{F}^{(01)} \bar{F}^{(01)} - 8(\xi + 2\zeta\eta) \xi^2 \tilde{\omega}_f \omega_f (\tilde{R} \cdot \bar{R}) \tilde{F}^{(10)} \bar{F}^{(10)} \\ &\quad - 8(\zeta + \xi)^2 m_f \xi \omega_f (\tilde{R} \cdot X) \tilde{F}^{(11)} \bar{F}^{(00)} + 8(1 - \zeta)^2 m_f \xi \tilde{\omega}_f (\bar{R} \cdot \bar{Y}) \tilde{F}^{(00)} \bar{F}^{(11)} \\ &\quad + 8\eta^2 m_f \xi \tilde{\omega}_f (\tilde{R} \cdot \bar{Y}) \tilde{F}^{(10)} \bar{F}^{(01)} - 8\zeta^2 m_f \xi \omega_f (X \cdot \bar{R}) \tilde{F}^{(01)} \bar{F}^{(10)} \end{aligned} \quad (3.31)$$

There is no $\tilde{F}^{(00)} \bar{F}^{(00)}$ term: the two double-flip paths do not interfere.

- Regular $\times$ Instantaneous + h.c.

$$\mathcal{H}_{\text{reg} \times \text{inst}}^\times = -\frac{8\xi\eta(\zeta + \xi)^2}{(1 - \zeta)} (\tilde{R} \cdot X) \tilde{F}^{(11)} \bar{\mathcal{G}}_E + \frac{8\xi\zeta(1 - \zeta)^2}{(\zeta + \xi)} (\bar{R} \cdot \bar{Y}) \mathcal{G}_E \bar{F}^{(11)} \quad (3.32)$$

using $\sum_{l,r} \tau_{\lambda_1 \lambda_1}^{mr}(\xi, \zeta) \varphi_{\lambda_1 \lambda_1}^{li}(\zeta + \xi)(\delta^{lr} + 2i\epsilon^{lr} \lambda_1)^* = 4(\zeta + \xi)^2(\delta^{im} - 2i\epsilon^{im} \lambda_1)$ and $\sum_{l,r}(\delta^{lr} - 2i\epsilon^{lr} \lambda_1)[\varphi_{\lambda_1 \lambda_1}^{lj}(\zeta)\tau_{\lambda_1 \lambda_1}^{rn}(\xi, \eta)]^* = -4(1 - \zeta)^2(\delta^{jn} - 2i\epsilon^{jn} \lambda_1)$. The mass terms do not interfere with the instantaneous terms here.

- Instantaneous $\times$ Instantaneous

$$\mathcal{H}_{\text{inst} \times \text{inst}}^\times = \frac{\zeta \xi}{(\zeta + \xi)} \frac{\xi \eta}{(1 - \zeta)} \mathcal{G}_E \bar{\mathcal{G}}_E \sum_{\lambda_1} \sum_{l,r} (\delta^{lr} - 2i\epsilon^{lr} \lambda_1) (\delta^{lr} + 2i\epsilon^{lr} \lambda_1)^* = \frac{\zeta \xi}{(\zeta + \xi)} \frac{\xi \eta}{(1 - \zeta)} \mathcal{G}_E \bar{\mathcal{G}}_E \sum_{\lambda_1} (2 - 8\lambda_1^2) = 0 \quad (3.33)$$

Massless limit: with $\frac{\tilde{R}^i X^m \bar{R}^j \bar{Y}^n}{X^2 \bar{Y}^2} \frac{Q^2 K_1 K_1}{\mathcal{D} \bar{\mathcal{D}}}$ taken out, Regular $\times$ Regular is the massless one and

$$\tilde{\Phi} \Phi^* \Big|_{\text{reg} \times \text{inst}} = -\frac{8\xi\eta(\zeta + \xi)^2}{(1 - \zeta)} \delta^{im} \delta^{jn} \frac{\bar{Y}^2}{\bar{R} \cdot \bar{Y}} + \frac{8\xi\zeta(1 - \zeta)^2}{(\zeta + \xi)} \delta^{im} \delta^{jn} \frac{X^2}{\tilde{R} \cdot X}, \quad \tilde{\Phi} \Phi^* \Big|_{\text{inst} \times \text{inst}} = 0.$$

$\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z})$ is the Wilson line structure (independent of the quark mass). This has 4 terms:

- $\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z})$

$$\mathcal{W}(x, y, z, \bar{x}, \bar{y}, \bar{z}) = S_{q\bar{q}g q\bar{q}g}(x, y, z, \bar{x}, \bar{y}, \bar{z}) - S_{q\bar{q}g}(x, y, z) - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) + 1 \quad (3.34)$$

where the Wilson line averaging is given by

$$\begin{aligned} S_{q\bar{q}g q\bar{q}g}(x, y, z, \bar{x}, \bar{y}, \bar{z}) &= \frac{1}{C_F N_c} \langle [U^\dagger(\bar{z})U(z)]^{ab} \text{Tr} [V(\bar{y})t^a V^\dagger(\bar{x})V(x)t^b V^\dagger(y)] \rangle \\ S_{q\bar{q}g}(x, y, z) &= \frac{1}{C_F N_c} \langle U^{ab}(z) \text{Tr} [V(x)t^a V^\dagger(y)t^b] \rangle \end{aligned}$$

At large N_c limit, we get

$$\begin{aligned} S_{q\bar{q}g q\bar{q}g}(x, y, z, \bar{x}, \bar{y}, \bar{z}) &= Q(x, z, \bar{z}, \bar{x}) Q(z, y, \bar{y}, \bar{z}) \\ S_{q\bar{q}g}(x, y, z) &= S(x, z) S(z, y) \\ S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) &= S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z}) \end{aligned}$$

Using Gaussian approximation,

$$Q(x, z, \bar{z}, \bar{x}) = S(x, z) S(\bar{z}, \bar{x}) - \frac{F(x, z; \bar{z}, \bar{x})}{F(x, \bar{z}; z, \bar{x})} [S(x, z) S(\bar{z}, \bar{x}) - S(x, \bar{x}) S(\bar{z}, z)] \quad (3.35)$$

$$Q(z, y, \bar{y}, \bar{z}) = S(z, y) S(\bar{y}, \bar{z}) - \frac{F(z, y; \bar{y}, \bar{z})}{F(z, \bar{y}; y, \bar{z})} [S(z, y) S(\bar{y}, \bar{z}) - S(z, \bar{z}) S(\bar{y}, y)] \quad (3.36)$$

- $\mathcal{W}(x', y, x', \bar{x}, \bar{y}, \bar{z})$

$$\mathcal{W}(x', y, x', \bar{x}, \bar{y}, \bar{z}) = S_{q\bar{q}g\ q\bar{q}g}(x', y, x', \bar{x}, \bar{y}, \bar{z}) - S_{q\bar{q}g}(x', y, x') - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) + 1 \quad (3.37)$$

where the Wilson line averaging is given by

$$S_{q\bar{q}g\ q\bar{q}g}(x', y, x', \bar{x}, \bar{y}, \bar{z}) = \frac{1}{C_F N_c} \langle [U^\dagger(\bar{z})U(x')]^{ab} \text{Tr} [V(\bar{y})t^a V^\dagger(\bar{x})V(x')t^b V^\dagger(y)] \rangle$$

$$S_{q\bar{q}g}(x', y, x') = \frac{1}{C_F N_c} \langle U^{ab}(x') \text{Tr} [V(x')t^a V^\dagger(y)t^b] \rangle$$

At large N_c limit, we get

$$S_{q\bar{q}g\ q\bar{q}g}(x', y, x', \bar{x}, \bar{y}, \bar{z}) = Q(x', x', \bar{z}, \bar{x}) Q(x', y, \bar{y}, \bar{z}) = S(\bar{z}, \bar{x}) Q(x', y, \bar{y}, \bar{z})$$

$$S_{q\bar{q}g}(x', y, x') = S(x', x') S(x', y) = S(x', y)$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}, \bar{z}) = S(\bar{z}, \bar{x}) S(\bar{y}, \bar{z})$$

Using Gaussian approximation,

$$Q(x', y, \bar{y}, \bar{z}) = S(x', y) S(\bar{y}, \bar{z}) - \frac{F(x', y; \bar{y}, \bar{z})}{F(x', \bar{y}; y, \bar{z})} [S(x', y) S(\bar{y}, \bar{z}) - S(x', \bar{z}) S(\bar{y}, y)] \quad (3.38)$$

- $\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}')$

$$\mathcal{W}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = S_{q\bar{q}g\ q\bar{q}g}(x, y, z, \bar{x}, \bar{y}', \bar{y}') - S_{q\bar{q}g}(x, y, z) - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') + 1 \quad (3.39)$$

where the Wilson line averaging is given by

$$S_{q\bar{q}g\ q\bar{q}g}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = \frac{1}{C_F N_c} \langle [U^\dagger(\bar{y}')U(z)]^{ab} \text{Tr} [V(\bar{y})t^a V^\dagger(\bar{x})V(x)t^b V^\dagger(y)] \rangle$$

$$S_{q\bar{q}g}(x, y, z) = \frac{1}{C_F N_c} \langle U^{ab}(z) \text{Tr} [V(x)t^a V^\dagger(y)t^b] \rangle$$

At large N_c limit, we get

$$S_{q\bar{q}g\ q\bar{q}g}(x, y, z, \bar{x}, \bar{y}', \bar{y}') = Q(x, z, \bar{y}', \bar{x}) Q(z, y, \bar{y}', \bar{y}') = Q(x, z, \bar{y}', \bar{x}) S(z, y)$$

$$S_{q\bar{q}g}(x, y, z) = S(x, z) S(z, y)$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) S(\bar{y}', \bar{y}') = S(\bar{y}', \bar{x})$$

Using Gaussian approximation,

$$Q(x, z, \bar{y}', \bar{x}) = S(x, z) S(\bar{y}', \bar{x}) - \frac{F(x, z; \bar{y}', \bar{x})}{F(x, \bar{y}'; z, \bar{x})} [S(x, z) S(\bar{y}', \bar{x}) - S(x, \bar{x}) S(\bar{y}', z)] \quad (3.40)$$

- $\mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}')$

$$\mathcal{W}(x', y, x', \bar{x}, \bar{y}', \bar{y}') = S_{q\bar{q}g\ q\bar{q}g}(x', y, x', \bar{x}, \bar{y}', \bar{y}') - S_{q\bar{q}g}(x', y, x') - S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') + 1 \quad (3.41)$$

where the Wilson line averaging is given by

$$S_{q\bar{q}g\ q\bar{q}g}(x', y, x', \bar{x}, \bar{y}', \bar{y}') = \frac{1}{C_F N_c} \langle [U^\dagger(\bar{y}')U(x')]^{ab} \text{Tr} [V(\bar{y}')t^a V^\dagger(\bar{x})V(x')t^b V^\dagger(y)] \rangle$$

$$S_{q\bar{q}g}(x', y, x') = \frac{1}{C_F N_c} \langle U^{ab}(x') \text{Tr} [V(x')t^a V^\dagger(y)t^b] \rangle$$

At large N_c limit, we get

$$S_{q\bar{q}g\ q\bar{q}g}(x', y, x', \bar{x}, \bar{y}', \bar{y}') = Q(x', x', \bar{y}', \bar{x}) Q(x', y, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) S(x', y)$$

$$S_{q\bar{q}g}(x', y, x') = S(x', x') S(x', y) = S(x', y)$$

$$S_{q\bar{q}g}^\dagger(\bar{x}, \bar{y}', \bar{y}') = S(\bar{y}', \bar{x}) S(\bar{y}', \bar{y}') = S(\bar{y}', \bar{x})$$

also

$$\mathcal{D}^2(x, y, z, \zeta, \xi) = \zeta (1 - \zeta) R^2 + \frac{\xi \eta Y^2}{(1 - \zeta)} = \zeta \eta (x - y)^2 + \zeta \xi (x - z)^2 + \eta \xi (y - z)^2$$

and

$$R = x - y'; \bar{R} = \bar{x} - \bar{y}'; Y = y - z; \bar{Y} = \bar{y} - \bar{z}; \tilde{R} = x' - y; X = x - z; \bar{X} = \bar{x} - \bar{z}; \tilde{\bar{R}} = \bar{x}' - \bar{y}$$

A Summary of the massive functions

All quantities for one flavour f ; $\eta = 1 - \zeta - \xi$, $a = \frac{\xi\eta}{\zeta(1-\zeta)^2}$, $\tilde{a} = \frac{\xi\zeta}{\eta(1-\eta)^2}$.

	antiquark emission	quark emission
photon scale	$\varepsilon_f^2 = \zeta(1-\zeta)Q^2 + m_f^2$	$\tilde{\varepsilon}_f^2 = \eta(1-\eta)Q^2 + m_f^2$
dead-cone scale	$\omega_f = \xi m_f / (1-\zeta)$	$\tilde{\omega}_f = \xi m_f / (1-\eta)$
three-parton scale	$E_f^2 = Q^2 + m_f^2(1/\zeta + 1/\eta)$	$\mathcal{D}^2 = \zeta\eta r_{xy}^2 + \zeta\xi r_{xz}^2 + \eta\xi r_{yz}^2$
before SW (regular)	$\mathcal{G}^{(\alpha\beta)}(R, Y)$, Eq. (2.10)	$\tilde{\mathcal{G}}^{(\alpha\beta)}(\tilde{R}, X)$: Eq. (2.10) with $\zeta \rightarrow \eta$, $a \rightarrow \tilde{a}$
after SW	$\mathcal{A}^{(\alpha\beta)}(R, Y)$, Eq. (2.17)	$\tilde{\mathcal{A}}^{(\alpha\beta)}(\tilde{R}, X)$
instantaneous	$\mathcal{G}_E = E_f K_1(E_f \mathcal{D}) / \mathcal{D}$ (the same function in both channels)	

The superscript $(\alpha\beta)$ records whether the photon vertex (α) and the gluon vertex (β) are helicity conserving (1, momentum in the numerator) or helicity flipping (0, mass in the numerator). Properties:

1. $m_f \rightarrow 0$: $\mathcal{G}^{(11)} \rightarrow \frac{1}{Y^2} \frac{Q K_1(Q\mathcal{D})}{\mathcal{D}}$, $\mathcal{A}^{(11)} \rightarrow \frac{1}{Y^2} \frac{Q^2 K_1(\tilde{Q}R)}{\tilde{Q}R}$, $\mathcal{G}_E \rightarrow \frac{Q K_1(Q\mathcal{D})}{\mathcal{D}}$; all other functions appear multiplied by m_f or $\omega_f \propto m_f$, so the massless note is recovered.
2. The regular functions are *not* a single Bessel function of a shifted scale: the two energy denominators have different mass shifts (ε_f and ω_f), and the Feynman parameter x interpolates between the two-body scale ($x \rightarrow 0$) and the three-body scale E_f ($x = 1$). Only the instantaneous term is exactly $Q \rightarrow E_f$.
3. $Y \rightarrow 0$ (gluon collinear to the antiquark): $Y^2 \mathcal{G}^{(11)}$ and $Y^2 \mathcal{A}^{(11)}$ tend to the same limit $\frac{\varepsilon_f K_1(\varepsilon_f R)}{\zeta(1-\zeta)R}$, $Y^2 \mathcal{G}^{(01)}$ and $Y^2 \mathcal{A}^{(01)}$ to $\frac{K_0(\varepsilon_f R)}{\zeta(1-\zeta)}$, and $\mathcal{G}^{(10)} - \mathcal{A}^{(10)}$, $\mathcal{G}^{(00)} - \mathcal{A}^{(00)}$ stay finite. Together with $\mathcal{W}(x, y, z, \dots) \rightarrow \mathcal{W}(x, y', y', \dots)$ this is the same ultraviolet cancellation as in the massless case.
4. Large distances are cut off by ε_f (and E_f) even at $Q^2 = 0$; the gluon emission after the shock wave is cut off at $Y \gtrsim 1/\omega_f$ (dead cone).

B Changes with respect to the original note

Line numbers refer to `Trijet_simplified.tex` as supplied. Items marked (M) are the mass changes; the others are corrections of the massless expressions (all of them are also needed to get the $m_f \rightarrow 0$ limit right).

1. (M) Everywhere: $\tilde{Q}^2 \rightarrow \varepsilon_f^2 = \tilde{Q}^2 + m_f^2$ in the photon energy denominator, $\tilde{p}^2 \rightarrow \tilde{p}^2 + \omega_f^2$ in the gluon-emission energy denominators, helicity-flip vertices $m_f \mu$ (photon) and $\xi \omega_f \mu$ (gluon) added, Q^2 prefactor of the cross sections moved into the radial functions, $\sum_f e_f^2$ moved inside the integrals, new functions $\mathcal{G}, \mathcal{A}, \mathcal{G}_E$ (Appendix A). The effective vertices $\Phi, \tilde{\Phi}$ (L196, L268) become the vertex functions $\mathbb{M}, \tilde{\mathbb{M}}$ (Eqs. (2.23), (2.35)) because regular and instantaneous pieces no longer share one radial function.
2. L68 vs. L71: taken literally, the bracket of L68 reduces to $[(2\zeta - 1)\delta^{\lambda j} - i\epsilon^{\lambda j} \sigma^3]$, not to the $+i\epsilon^{\lambda j} \sigma^3$ of L71–L74. The reduced form of L71–L74 (and φ of L87) is the one consistent with the gluon vertex L78 (τ of L87) and with the Dirac-spinor amplitude (Appendix C), so it is kept, and the σ matrices in the bracket are written in the order that reproduces it: $-\sigma^\lambda \frac{\sigma \cdot p}{p^+} - \frac{\sigma \cdot k}{k^+} \sigma^\lambda$.
3. L71, L74: $(p^j - \zeta q^j) \rightarrow (k^j - \zeta q^j) = \tilde{k}^j$ (k is the quark momentum, ζ its fraction).
4. L138: the exponent $e^{+i\tilde{p} \cdot Y}$ is inconsistent with the phase $e^{-i\tilde{p} \cdot Y}$ derived in L118; with $e^{-i\tilde{p} \cdot Y}$ the right-hand side has a + sign. The final massless Eq. (eq:1) of the note (L146, overall $-$) is consistent with the corrected form, so it is unchanged. The generic formula of L134 is correct as written (with $e^{+i\tilde{p} \cdot Z}$).
5. L140: “is is” $\rightarrow$ “is”.
6. L159, L243: energy denominator $E_q(k') + E_{\bar{q}}(p') - E_q(k'') + E_{\bar{q}}(p'') - E_g(r) \rightarrow E_q(k') + E_{\bar{q}}(p') - E_q(k'') - E_{\bar{q}}(p'') - E_g(r)$.
7. L164: $\zeta(1-\zeta)\tilde{p}^2 \rightarrow \zeta(1-\zeta)^2\tilde{p}^2$ (on both sides of the arrow).
8. L176: “Instantaneous” $\rightarrow$ “Instantaneous”.
9. L187: the denominator $\left[\tilde{k}^2 + \tilde{Q}^2 + \frac{\zeta(1-\zeta)^2}{\xi\eta}\right]\tilde{p}^2$ should be $\tilde{k}^2 + \tilde{Q}^2 + \frac{\zeta(1-\zeta)^2}{\xi\eta}\tilde{p}^2$ (a single three-parton denominator). As printed, the overall sign, the power of q^+ and a factor $2\zeta\xi\eta\sqrt{q^+}$ are also inconsistent with L191; the momentum-space expression was rewritten so that its Fourier transform gives L191 (L191, and hence Φ , are correct: their relative sign and weight with respect to the regular term were checked against the Dirac-spinor amplitude).
10. L183–L191, L256–L260: Fock state written as $|q\bar{q}g\rangle$ in the same order as the regular terms (notation only).
11. L193: the sentence “For, the counterpart ...” rephrased (with masses there is no $\frac{Y^2}{R \cdot Y}$ factor; there is simply no instantaneous term after the shock wave).
12. L219: “=” $\rightarrow$ “ $\rightarrow$ ” in the rule for φ .

13. L222, L247: in the quark-emission channel the photon scale is $\tilde{Q}_q^2 = \eta(1-\eta)Q^2$ (the photon splits into a quark with fraction $1-\eta$ and the antiquark with fraction η), not $\tilde{Q}^2 = \zeta(1-\zeta)Q^2$; massive: $\tilde{\epsilon}_f$.
14. L289 (antiquark, Regular $\times$ Instantaneous): corrected to $-\frac{8\zeta\xi\eta^2}{(1-\zeta)}\delta^{im}\delta^{jn}\left[\frac{Y^2}{R\cdot Y} + \frac{\bar{Y}^2}{R\cdot\bar{Y}}\right]$. The printed tensor $\zeta(2(1-\zeta) - \xi)\delta^{in}\delta^{jm} - \xi(1-\zeta)\epsilon^{in}\epsilon^{jm}$ cannot be right: the instantaneous term carries δ^{nj} between its two coordinate indices, so the result must be proportional to δ^{im} (resp. δ^{jn}); the coefficient follows from the identity quoted below Eq. (3.4).
15. L293 (antiquark, Instantaneous $\times$ Instantaneous): $8\delta^{ij}\delta^{nm}\left[\frac{\xi\zeta}{1-\zeta}\right]^2 \rightarrow 8\delta^{im}\delta^{jn}\left[\frac{\xi\eta}{1-\zeta}\right]^2$ (the instantaneous coefficient in Φ is $\frac{\xi\eta}{1-\zeta}$, and $\sum_{\lambda_1, l, r} |\delta^{lr} + 2i\epsilon^{lr}\lambda_1|^2 = 8$).
16. L303, L344, L359, L399, L432, L447, L488, L521: missing “=” in the definition of $S_{q\bar{q}g\,q\bar{q}g}$.
17. L313–L314: the same line was printed twice; one copy removed.
18. L377: $\tilde{\Phi}\Phi^* \rightarrow \tilde{\Phi}\tilde{\Phi}^*$ for the quark direct term.
19. L385 (quark, Regular $\times$ Instantaneous): corrected to $+\frac{8\zeta^2\xi\eta}{(1-\eta)}\delta^{im}\delta^{jn}\left[\frac{X^2}{R\cdot X} + \frac{\bar{X}^2}{R\cdot\bar{X}}\right]$ (same reason as L289; the overall sign differs from the antiquark case because $R \rightarrow -\tilde{R}$).
20. L389 (quark, Instantaneous $\times$ Instantaneous): $\left[\frac{\xi\eta}{1-\eta}\right]^2 \rightarrow \left[\frac{\xi\zeta}{1-\eta}\right]^2$ and $\delta^{ij}\delta^{nm} \rightarrow \delta^{im}\delta^{jn}$.
21. L474 (interference, Regular $\times$ Instantaneous): corrected to $-\frac{8\xi\eta(\zeta+\xi)^2}{(1-\zeta)}\delta^{im}\delta^{jn}\frac{\bar{Y}^2}{R\cdot Y} + \frac{8\xi\zeta(1-\zeta)^2}{(\zeta+\xi)}\delta^{im}\delta^{jn}\frac{X^2}{R\cdot X}$.
22. L478 (interference, Instantaneous $\times$ Instantaneous): the intermediate expression was missing the helicities, $(\delta^{li} - 2i\epsilon^{li}) \rightarrow (\delta^{lr} - 2i\epsilon^{lr}\lambda_1)$ etc.; the result 0 is correct.
23. Unchanged and verified: the Regular $\times$ Regular expressions of L285, L381, L470 (including the signs of the $\epsilon\epsilon$ terms), the relative signs and weights of regular, after-shock-wave and instantaneous terms in Φ and $\tilde{\Phi}$, the relative sign between quark and antiquark emission (overall $-\eta\zeta$ of the interference), and all Wilson-line structures.

This note supersedes the earlier `trijet_massive_quarks.pdf`, which replaced $QK_1(Q\mathcal{D}) \rightarrow E_f K_1(E_f\mathcal{D})$ in *all* terms and omitted the helicity-flip terms; that replacement is exact only for the instantaneous term.

C How the results were checked

The scripts are in `trijet/checks/massive/` of the repository.

1. **Vertices** (`checkE.vertices.py`): the products $\sum_{\lambda'} [\bar{u}\not{\epsilon}_\gamma v][\bar{v}\not{\epsilon}_g v]$ and $\sum_{\lambda'} [\bar{u}\not{\epsilon}_g u][\bar{u}\not{\epsilon}_\gamma v]$ were computed with massive Lepage–Brodsky spinors for all helicities and polarisations and fitted, as polynomials in $\tilde{k}, \tilde{p}, m_f$, to $c\sum_{\lambda}(\varphi\tilde{k} + m_f\mu)(\tau\tilde{p} + \xi\omega_f\mu)$ and to the quark-line analogue. The fit has a unique solution, the m_f^2 terms then match with no freedom left, and the same constant c appears in both channels.
2. **Complete amplitude** (`checkF_full_amplitude.py`): the full amplitude $\bar{u}(k_1)[\not{\epsilon}_g\frac{K+m}{K^2-m^2}\not{\epsilon}_\gamma + \not{\epsilon}_\gamma\frac{-\bar{P}+m}{\bar{P}^2-m^2}\not{\epsilon}_g]v(k_2)$ (which contains the regular and the instantaneous terms of both emitters) equals $q^+\sqrt{\zeta\eta}$ times the momentum-space two-component expressions of Sec. 2 (regular + instantaneous, quark + antiquark emission, with the signs used above), exactly, for all 16 helicity/polarisation combinations, several momenta, masses (including $m_f = 0$) and momentum fractions.
3. **Fourier transforms** (`checkG.fourier.py`): Eq. (2.10) agrees with an independent evaluation (exact $\tilde{p}$ integral followed by a one-dimensional Hankel transform in $\tilde{k}$) to 10^{-13} ; $\mathcal{G}^{(11)}$ reduces to $\frac{1}{Y^2}\frac{QK_1(Q\mathcal{D})}{D}$ at $m_f = 0$; Eq. (2.21) holds; the $Y \rightarrow 0$ limits of Appendix A hold.
4. **Helicity sums** (`checkH_helicity_sums.py`): all $\mathcal{H}, \tilde{\mathcal{H}}, \mathcal{H}^\times$ of Sec. 3 were obtained by summing the vertex functions over $\lambda_1, \lambda_2, l, r$ symbolically in ζ, ξ, m_f and decomposing the result on the rotation-invariant tensors; the massless Regular $\times$ Regular expressions of the original note are reproduced.